

Cisco

CCNP Enterprise

Cisco Certified Network Professional – Enterprise

300 Interview Questions & Answers



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How to use this guide: Each question is explained in clear, user-friendly language with real-world analogies. Green TIP boxes flag exam-critical points. All 10 sections cover CCNP Enterprise 350-401 ENCOR + 300-410 ENARSI syllabus. Prepared by Akhilesh Chaudhari – Senior IT Engineer.

SECTION 01

Advanced Routing — OSPF & EIGRP

Q1 – Q30

OSPFv2/v3 · Multi-Area · EIGRP Advanced · Redistribution · Path Selection · Tuning

Q1 What is OSPF and how does it build its routing table?

Ans OSPF (Open Shortest Path First) is a link-state routing protocol — every router builds a complete map of the network. Process: 1) Routers form neighbour adjacencies (exchange Hello packets). 2) They flood Link State Advertisements (LSAs) describing their connections to all neighbours. 3) Every router builds an identical Link State Database (LSDB). 4) Each router runs Dijkstra's SPF algorithm on the LSDB to find shortest path to every destination. 5) Best paths go into routing table.

TIP CCNP ENCOR = 350-401. OSPF is 15-20% of the exam. Master every detail!

Q2 What are the different OSPF LSA types?

Ans LSA types and what they carry: Type 1 (Router LSA): every router generates this — describes its own directly connected links. Stays within area. Type 2 (Network LSA): generated by DR on multi-access networks (Ethernet). Stays within area. Type 3 (Summary LSA): ABR generates these to summarise routes between areas. Type 4 (ASBR Summary): tells routers how to reach an ASBR. Type 5 (External LSA): ASBR generates — external routes (from redistribution). Floods entire OSPF domain. Type 7 (NSSA External): external routes allowed into NSSA areas.

TIP LSA types 1-7 are directly tested in CCNP. Know which router generates each type!

Q3 What is OSPF area types — Stub, Totally Stub, NSSA?

Ans Regular area: receives all LSA types including Type 5 external routes. Stub area: blocks Type 5 external LSAs — router replaced with default route. Reduces LSDB size. Totally Stub area (Cisco proprietary): blocks Type 3, 4, and 5. Only a default route from ABR. Smallest LSDB possible — best for remote branches. NSSA (Not So Stubby Area): allows limited external routes (redistributed as Type 7 LSAs) but still blocks Type 5. ABR converts Type 7 to Type 5 before entering backbone. Use when redistributing into an area that should otherwise be stub.

Q4 What is the difference between OSPF E1 and E2 external routes?

Ans External routes redistributed into OSPF are either E1 or E2. E2 (default): metric = external metric only (set at redistribution point — does NOT increase as route travels through OSPF). E1: metric = external metric + internal OSPF cost to reach ASBR (increases hop by hop). When should you use E1? When you have multiple ASBRs redistributing the same external routes — E1 ensures routers use the closer ASBR. E2 is fine when only one ASBR exists. Configure: redistribute ospf 1 subnets metric-type 1.

Q5 What is OSPF virtual link and when is it needed?

Ans OSPF requires all areas to connect directly to Area 0 (backbone). Virtual link creates a logical connection through a non-backbone area to connect a disconnected area to Area 0. Example: Area 2 connects to Area 1, and Area 1 connects to Area 0 — but Area 2 cannot reach Area 0 directly. Create virtual link through Area 1: area 1 virtual-link [Router-ID of far ABR]. Both transit area ABRs must configure the virtual link. This is a temporary fix — better to redesign topology.

Q6 What is OSPF summarisation and how to configure it?

Ans Route summarisation reduces the number of routes advertised, shrinking routing tables and speeding up convergence. ABR summarisation (between areas): area 1 range 192.168.0.0 255.255.0.0 — summarises all /24s in area 1 into one /16 advertised to backbone. ASBR summarisation (external routes): summary-address 10.0.0.0 255.0.0.0 — summarises redistributed external routes. Benefits: smaller routing tables, faster SPF calculation, reduces flapping impact (one subnet going down no longer triggers full SPF across domain).

Q7 What is OSPFv3 and how does it differ from OSPFv2?

Ans OSPFv3 is OSPF for IPv6. Key differences: Runs over IPv6 (link-local addresses for neighbour discovery), configured on interface (not network statement), supports IPv6 address families. New feature: address family model — OSPFv3 can carry both IPv4 and IPv6 routes simultaneously using address families: ipv6 router ospf vs router ospfv3. Process ID stays same. LSA structure updated (Type 8 and 9 added). Authentication moved to IPv6 IPsec. Configure: ipv6 ospf 1 area 0 on interface.

Q8 How does OSPF DR/BDR election work and why does it matter?

Ans On Ethernet segments with multiple OSPF routers, all routers form full adjacency only with the DR (Designated Router) and BDR (Backup DR) — not with every router. This dramatically reduces OSPF traffic ($n-1$ vs $n*(n-1)/2$ adjacencies). Election: highest priority wins (ip ospf priority 255 = definite DR). Tie: highest Router ID wins. Important: election is non-preemptive — if DR fails, BDR becomes DR and new BDR elected. To force a specific router as DR: set its priority highest AND force re-election by clearing OSPF process.

Q9 What is OSPF network type and why does it matter?

Ans OSPF network type controls whether DR/BDR election happens and hello/dead timer values. Types: Broadcast (default Ethernet): DR/BDR election, hello 10s/dead 40s. Point-to-Point (serial, tunnel): no DR/BDR, hello 10s/dead 40s. Non-Broadcast (Frame Relay, ATM): DR/BDR elected, hello 30s/dead 120s — neighbours configured manually. Point-to-Multipoint: no DR/BDR, hello 30s/dead 120s — auto discovers neighbours. Mismatch in network type between neighbours prevents full adjacency — both sides must match.

TIP *Network type mismatch = neighbours stuck in 2-WAY state. Classic CCNP exam scenario!*

Q10 What is EIGRP and how is its metric calculated?

Ans EIGRP (Enhanced Interior Gateway Routing Protocol) uses a composite metric. Default formula: $\text{Metric} = 256 * (10^7 / \text{Min-Bandwidth} + \text{Cumulative-Delay})$. Bandwidth: minimum bandwidth along the entire path (in Kbps). Delay: sum of delays on all interfaces in the path (in tens of microseconds). K-values control which metrics are included (default: K1=1, K2=0, K3=1, K4=0, K5=0 — only bandwidth and delay matter by default). All routers in an EIGRP domain must have identical K-values — mismatch prevents adjacency.

Q11 What is EIGRP DUAL algorithm?

Ans DUAL (Diffusing Update Algorithm) is what makes EIGRP fast and loop-free. Key concepts: Feasible Distance (FD): best metric from this router to destination. Reported Distance (RD): metric neighbour says it costs them to reach destination. Successor: best route (lowest FD) — goes in routing table. Feasible Successor (FS): backup route — RD must be less than current FD (Feasibility Condition). When Successor fails: EIGRP instantly switches to FS — no recalculation needed. No FS: EIGRP goes Active and queries all neighbours.

Q12 What is EIGRP Named Mode vs Classic Mode?

Ans Classic mode (older): `router eigrp 1` □ network statements □ manual configuration per feature. Named mode (IOS 15.0+): `router eigrp MY-INSTANCE` □ address-family ipv4 unicast autonomous-system 1 □ all config in one place. Named mode benefits: supports IPv4 and IPv6 in same process, better organisation, additional features (wide metrics — supports higher than 4Gbps), consistent configuration model. Show differences: `show eigrp address-family ipv4 topology`. Most enterprise networks should use named mode for new deployments.

Q13 What is EIGRP stub routing and when should you use it?

Ans EIGRP stub tells the hub router 'this spoke router does not have any downstream EIGRP neighbours — do not ask it for routes during an Active query'. Benefits: prevents query propagation to stub routers (reduces query domain), faster convergence during failures in the hub network. Configure on spoke: `eigrp stub connected summary`. The stub router still receives and uses routes from hub — it just cannot be a transit. Use in all hub-and-spoke topologies (Frame Relay, DMVPN, SD-WAN with EIGRP).

Q14 How does EIGRP unequal-cost load balancing work?

Ans EIGRP supports load balancing across paths with different metrics (unlike OSPF which only supports equal-cost). The `variance` command controls this. Variance N means: include routes with FD up to N times the best FD in load balancing. Example: best FD = 1000, variance 2 = routes with FD up to 2000 are used. Traffic distribution is proportional to metric (more traffic on better path). Requirement: the route must still pass Feasibility Condition to be used. Verify: `show ip eigrp topology` (routes marked as Successors and Feasible Successors).

Q15 What is route redistribution and what are the risks?

Ans Route redistribution imports routes from one routing protocol into another. Common scenarios: OSPF into EIGRP, static routes into BGP, connected routes into OSPF. Risks: routing loops (mutual redistribution — routes bounce between protocols), suboptimal routing (metrics reset at redistribution point), metric mismatch (EIGRP needs metric when importing non-EIGRP routes). Mitigations: use route tags to track redistributed routes and prevent loops, use route maps to filter what gets redistributed, redistribute in one direction only when possible.

Q16 What is route tagging in redistribution?

Ans Route tags attach a numeric value to routes that persists as they are redistributed. Use for preventing redistribution loops in mutual redistribution scenarios. Strategy: when redistributing OSPF→EIGRP, tag those routes with 100. When redistributing EIGRP→OSPF, deny routes with tag 100 (these came from OSPF originally — do not send back). Configuration: redistribute ospf 1 subnets route-map OSPF-TO-EIGRP; route-map OSPF-TO-EIGRP: deny 10 match tag 100; permit 20 set tag 100. Prevents routes looping back to their origin protocol.

Q17 What is OSPF authentication configuration?

Ans OSPF authentication prevents unauthorised routers from joining your OSPF domain. Two types: MD5 authentication (recommended — hash of password): ip ospf authentication message-digest on interface + ip ospf message-digest-key 1 md5 MyKey. Plain text (avoid in production — insecure): ip ospf authentication + ip ospf authentication-key MyPass. Area-wide: area 0 authentication message-digest. Both routers must match key ID and key value. Authentication failure = neighbours stay in INIT state, never reaching Full. OSPFv3 uses IPv6 IPsec instead.

Q18 What is EIGRP authentication?

Ans EIGRP authentication uses MD5 to verify neighbour identity. Configure key chain: key chain EIGRP-KEYS □ key 1 □ key-string MyPassword. Apply to interface: ip authentication mode eigrp 1 md5 + ip authentication key-chain eigrp 1 EIGRP-KEYS. Both routers need matching key chain and key-string. Key rollover: add new key with future accept and send times, then remove old key — graceful key rotation without disruption. EIGRP Named Mode: authenticate □ md5 or hmac-sha-256. HMAC-SHA-256 is more secure than MD5.

Q19 What is OSPF demand circuit?

Ans OSPF demand circuit suppresses periodic Hello and LSA refresh flooding on low-bandwidth or metered links (like ISDN, cellular backup links). Instead of sending Hellos every 10 seconds continuously, OSPF only sends updates when the topology changes. The interfaces are marked as demand circuits: ip ospf demand-circuit. The DN (Do Not Age) bit is set in LSAs on demand circuits — LSAs do not age out. Useful for reducing call charges on legacy ISDN dial-up backup links or minimising cellular data usage.

Q20 What is OSPF graceful restart?

Ans OSPF graceful restart allows a router restarting its OSPF process to maintain routing continuity. The restarting router signals its neighbours: 'I am restarting — please hold my routes for a grace period.' Neighbours enter helper mode and do not withdraw routes pointing through the restarting router. The restarting router rebuilds its LSDB and routing table during the grace period without causing traffic disruption. Configure: `nsf cisco` (Cisco NSF) or `nsf ietf` (standards-based). Requires neighbouring routers to support helper mode.

Q21 What is EIGRP wide metrics?

Ans Classic EIGRP metrics cannot represent bandwidth above 1 Gbps (the metric formula tops out). Wide metrics (EIGRP Named Mode) use 64-bit values supporting up to 655 Tbps. The metric divisor changes from 256 to 65536. Wide metrics also add two new K-values: K6 (jitter) and K7 (load). Wide metrics are automatically used when Named Mode EIGRP is configured. Not backward compatible with Classic EIGRP (different metrics). Verify: `show eigrp address-family ipv4 topology` (see if wide metrics active).

Q22 What is OSPF path selection when multiple paths exist?

Ans OSPF path selection order: 1) Intra-area routes (O) preferred over inter-area (O IA) preferred over external type 1 (O E1) preferred over external type 2 (O E2). 2) Within same type: lowest cost wins. 3) Equal cost paths: load balance (up to 16 equal-cost paths in IOS). To influence path: change interface cost (`ip ospf cost 100`), change reference bandwidth (`auto-cost reference-bandwidth 10000`), or use summarisation. Never change ospf priority to influence routing — priority only affects DR election.

Q23 What is EIGRP auto-summary and why should it be disabled?

Ans EIGRP auto-summary automatically summarises routes to their classful boundary when an interface crosses a classful boundary. Problem: in modern networks with VLSM (variable-length subnets), auto-summary creates black holes. Example: router has 172.16.1.0/24 and 172.16.2.0/24 subnets. Auto-summary advertises 172.16.0.0/16 — hides the detail. Remote router sends all 172.16.x.x traffic this way, even if some subnets are unreachable. Fix: `no auto-summary` under `router eigrip`. Always disable in modern networks.

Q24 What is OSPF BFD integration?

Ans BFD (Bidirectional Forwarding Detection) dramatically speeds up OSPF failure detection. Without BFD: OSPF dead timer = 40 seconds before declaring neighbour down. With BFD: detect failure in 50-300 milliseconds. How: BFD runs a separate fast hello mechanism independent of OSPF. When BFD detects failure: immediately notifies OSPF. OSPF removes the route and reconverges within seconds total. Configure: `bfd interval 200 min_rx 200 multiplier 3` on interface + `ip ospf bfd` on interface. Verify: `show bfd neighbors`.

Q25 What is EIGRP passive interface?

Ans `passive-interface [interface]` stops EIGRP from sending Hello packets out that interface while still advertising the connected network. Use on interfaces connected to end devices (PCs, printers) that do not run EIGRP — wastes bandwidth sending Hellos to devices that cannot respond. Also a security measure: no EIGRP neighbour can form on that interface — prevents rogue EIGRP neighbours. Configure: `passive-interface GigabitEthernet0/1`. Or make all interfaces passive by default: `passive-interface default` then activate specific interfaces: `no passive-interface Gi0/0`.

Q26 What is OSPF sham-link?

Ans In MPLS L3VPN networks, customer sites often have a backdoor link in addition to the VPN. OSPF sees the backdoor as an intra-area path (lower preference) while the MPLS VPN path appears as an inter-area path — OSPF prefers the backdoor even if the VPN path is better. A sham-link is a virtual intra-area link created within the VPN overlay, making the VPN path look like an intra-area path too. This allows normal OSPF cost comparison. Configure on PE routers: `area 0 sham-link [source-lo] [dest-lo] cost 10`.

Q27 What is EIGRP route filtering?

Ans EIGRP route filtering controls which routes are sent to or received from neighbours. Methods: `distribute-list` (uses ACL or prefix-list): `distribute-list ACL-NAME in/out [interface]`. Route map: more granular — match and set actions. Offset-list: add offset to metric of specific routes. Filtering is essential for: preventing certain routes from being advertised to specific neighbours (policy routing), controlling which routes a stub router receives, preventing suboptimal routing. Always filter inbound at the edge — do not accept routes you do not expect.

Q28 What is OSPF incremental SPF (iSPF)?

Ans Standard SPF recalculates the entire shortest path tree whenever any LSA changes — CPU intensive in large networks. Incremental SPF (iSPF) only recalculates the affected portion of the tree. For example: if a Type 1 LSA changes in one area, only that area's SPF is recalculated — not the entire LSDB. Configure: `ispf` (under router `ospf`). Benefits: faster convergence, lower CPU usage during topology changes. Most relevant in large networks with hundreds of routers. Verify: `show ip ospf statistics` (see SPF calculations).

Q29 What is EIGRP SIA (Stuck in Active) state?

Ans When EIGRP has no Successor and no Feasible Successor for a route, it goes Active and sends Query packets to all neighbours asking for alternative paths. If a neighbour does not reply within the Active timer (3 minutes default), the route gets Stuck in Active (SIA). Consequence: EIGRP resets the neighbour relationship — disruptive. Prevention: EIGRP stub (stub routers do not forward queries — reduces query domain), summary routes (queries stop at summarisation boundary), tune Active timer: `timers active-time 1` (reduce to 1 minute).

Q30 What is OSPF fast convergence tuning?

Ans Tune OSPF for faster convergence: Reduce Hello/Dead timers (`ip ospf hello-interval 1`, `ip ospf dead-interval 3` — but both sides must match). BFD (sub-second failure detection). SPF throttle (`timers throttle spf 50 200 5000` — first SPF after 50ms, max delay 5 seconds between SPF runs). LSA throttle (`timers throttle lsa 50 200 5000` — control LSA generation rate). Incremental SPF (`ispf`). All these together achieve sub-second OSPF convergence in well-designed networks.

SECTION 02	BGP Advanced	Q31 – Q60
eBGP · iBGP · Path Attributes · Policies · Route Reflectors · Filtering · Troubleshooting		
Q31	What is BGP and why is it the internet's routing protocol?	
Ans	BGP (Border Gateway Protocol) routes traffic between different organisations (Autonomous Systems – AS). It is the only EGP (Exterior Gateway Protocol) in use today. Why BGP? It scales to carry over 900,000 routes (the full internet routing table), supports complex policy-based routing (not just shortest path), provides path-loop prevention (AS-path), and allows fine-grained traffic engineering. Every ISP, large enterprise with multiple providers, and cloud provider uses BGP. CCNP requires deep understanding of BGP path selection and policies.	
TIP	BGP is 15-20% of CCNP ENCOR. Understand path selection and attributes deeply!	
Q32	What are the BGP path selection steps in order?	
Ans	BGP selects best path using this ordered list – first criterion that eliminates options wins: 1) Highest Weight (Cisco proprietary, local to router). 2) Highest Local Preference (iBGP only, influences outbound traffic). 3) Locally originated routes (network command or redistribute). 4) Shortest AS-path length. 5) Lowest Origin code (IGP < EGP < Incomplete). 6) Lowest MED (only compared between routes from same AS). 7) eBGP over iBGP. 8) Lowest IGP metric to BGP next-hop. 9) Oldest eBGP route. 10) Lowest Router ID. 11) Lowest neighbour IP address.	
TIP	Memorise BGP path selection order – it appears directly in CCNP exam questions!	
Q33	What is BGP Weight and how do you set it?	
Ans	Weight is a Cisco-proprietary BGP attribute – local to the router only (not advertised to any peer). Range: 0-65535. Higher weight = preferred. Default: routes from local network commands have weight 32768; routes received from BGP neighbours have weight 0. Set weight for a specific neighbour: neighbor 10.0.0.1 weight 100 (all routes from that neighbour get weight 100). Or per route with route-map: match ip address prefix-list MY-LIST then set weight 200. Use Weight to prefer one eBGP peer over another for outbound traffic on a single router.	

Q34 What is Local Preference and how does it work?

Ans Local Preference (LP) is shared only within your AS (iBGP) — not sent to eBGP peers. Higher LP = preferred outbound exit from your AS. Default: 100. Set LP: `neighbor 10.0.0.2 route-map SET-LP in; route-map SET-LP: set local-preference 200`. When BGP router receives same prefix from two eBGP peers: set higher LP for preferred exit → all iBGP routers prefer that exit. Use case: you have two ISPs — set LP 200 for ISP-A routes and LP 100 for ISP-B routes → all internal traffic exits via ISP-A by default.

Q35 What is MED (Multi-Exit Discriminator)?

Ans MED (metric attribute) influences inbound traffic from your neighbouring AS — you suggest which of your entry points they should use. Lower MED = preferred. MED is only compared between routes from the SAME neighbouring AS (by default). Set MED: `set metric [value]` in route-map applied outbound to eBGP peer. Example: you have two routers connecting to same ISP — set lower MED on router you want ISP to prefer for sending traffic to you. Note: ISPs are not required to honour MED values — it is advisory only.

Q36 What is AS-path prepending and why use it?

Ans AS-path prepending artificially makes your BGP path look longer (less preferred) by adding extra copies of your own AS number to the AS-path. Use case: you have two ISPs and want ISP-B to be backup only. To ISP-B, prepend your AS number 3 times: `set as-path prepend 65001 65001 65001` in route-map applied outbound to ISP-B peer. BGP shorter AS-path wins — ISP-B now sees a longer path through your network and prefers ISP-A for traffic destined to you. Simple and widely used for inbound traffic engineering.

Q37 What is iBGP and the full-mesh requirement?

Ans iBGP (internal BGP) sessions between routers within the same AS distribute BGP routes internally. Critical rule: routes learned via iBGP are NOT re-advertised to other iBGP peers — prevents routing loops. This means all iBGP routers must peer with each other (full mesh). Problem: with 10 routers, that is $10 \times (10-1) / 2 = 45$ sessions. With 100 routers: 4950 sessions — clearly unscalable. Solutions: Route Reflectors (eliminate full mesh — most common) or BGP Confederations (divide AS into sub-ASes).

Q38 What is a BGP Route Reflector?

Ans A Route Reflector (RR) eliminates the iBGP full mesh by acting as a central point. RR clients only peer with the RR — not with each other. The RR re-advertises iBGP routes to its clients (which normal iBGP rules prohibit). RR adds attributes: Originator-ID (prevents route being reflected back to originating router) and Cluster-list (prevents routing loops between multiple RRs). Configure: neighbor [client-IP] route-reflector-client. For redundancy: deploy two RRs (same cluster ID: bgp cluster-id [ID]). Most common iBGP scaling solution.

Q39 What is BGP confederation?

Ans BGP confederation divides a large AS into smaller sub-ASes. Internal sub-ASes use private AS numbers (64512-65534). External sub-ASes use the real public AS number. eBGP runs between sub-ASes (routes re-advertised freely) but appears as iBGP to external peers (AS-path shows only the public AS). Benefits: full mesh only required within each small sub-AS. Drawbacks: complex to configure and manage, AS-path manipulation required. Route Reflectors are simpler — confederations used in very large carrier networks (common in older SP deployments).

Q40 What is BGP next-hop issue and the next-hop-self solution?

Ans When an eBGP router receives a route, the next-hop IP is the eBGP peer's address. When this router advertises the route to iBGP peers, the next-hop IP stays unchanged (eBGP peer address). Problem: iBGP peers cannot reach this eBGP peer address because it is on an external link — BGP route blackholed. Solution: neighbor [iBGP-peer] next-hop-self — the iBGP-advertising router replaces next-hop with its own address. Alternative: redistribute the eBGP peer's subnet into IGP (OSPF/EIGRP) so iBGP peers can reach it.

Q41 What is BGP community attribute?

Ans BGP communities are tags attached to routes that carry additional information. Format: AS:value (e.g. 65001:100). Well-known communities: no-export (do not advertise outside AS), no-advertise (do not advertise to any peer), local-AS (do not advertise outside confederation sub-AS). Custom communities enable complex policies: ISP might accept customer's community 65100:200 to set lower LP on their end. Send communities: neighbor [IP] send-community. Without this, communities are not sent. Match in route-map: match community COMM-LIST.

Q42 What is BGP route filtering with prefix lists?

Ans Prefix lists are the preferred method for BGP route filtering (more efficient than ACLs). `ip prefix-list MY-LIST seq 5 permit 192.168.0.0/22 le 24` (permit any subnet within 192.168.0.0/22 with mask /22 to /24). Apply inbound or outbound: `neighbor [IP] prefix-list MY-LIST in`. Best practice: always filter BGP routes at the edge. What to filter inbound from ISP: deny default route (you should know your own default), deny your own prefixes (prevent route hijacking), permit legitimate prefixes only.

Q43 What is BGP soft reset?

Ans Hard reset (`clear ip bgp [IP]`): tears down TCP session, re-establishes BGP, re-exchanges entire routing table — causes brief traffic loss. Soft reset: re-applies BGP policies without resetting the TCP session. Two types: Inbound soft reset (`clear ip bgp [IP] soft in`): re-processes received routes through new inbound policies — requires route-refresh capability (RFC 2918) or stored received routes (`neighbor [IP] soft-reconfiguration inbound`). Outbound soft reset (`clear ip bgp [IP] soft out`): re-advertises outbound routes with current policies — no disruption. Always use soft reset in production.

Q44 What is BGP peer groups?

Ans Peer groups simplify BGP configuration when multiple neighbours share the same policies. Create: `neighbor UPSTREAM peer-group`. Configure group: `neighbor UPSTREAM remote-as 65002`, `neighbor UPSTREAM route-map SET-LP in`. Add members: `neighbor 10.0.0.1 peer-group UPSTREAM`, `neighbor 10.0.0.2 peer-group UPSTREAM`. All group members inherit group configuration. Performance: IOS processes BGP updates once for the group (not per-member) — significant CPU savings with many peers. Also simplifies configuration changes — update group config, all members updated.

Q45 What is 4-byte AS number in BGP?

Ans Original BGP used 2-byte AS numbers (1-65535). Internet growth exhausted these. 4-byte ASNs (RFC 4893) extend to 4,294,967,295. New format: plain number (131072) or dotted notation (2.0). Old routers send `AS_TRANS` (23456) for 4-byte ASNs they cannot understand — backward compatible. IOS configures: `router bgp 131072` or `router bgp 2.0`. Private 4-byte ASN range: 4200000000-4294967294. Check current ASN of internet peers: `show bgp ipv4 unicast summary`. Important for ISPs and large enterprises receiving new AS assignments.

Q46 What is BGP multipath?

Ans BGP installs only ONE best path per prefix in the routing table by default. Multipath allows multiple equal-cost BGP paths to be installed simultaneously for load balancing. iBGP multipath: maximum-paths ibgp 4 (up to 4 iBGP equal-cost paths). eBGP multipath: maximum-paths 4. Conditions for eBGP multipath: same AS-path length, same MED, same local-preference, different next-hops. Common in data centre spine routers receiving same prefix from multiple BGP peers. Verify: show bgp ipv4 unicast 192.168.1.0 (see multiple paths marked as multipath).

Q47 What is BGP dampening?

Ans Route dampening penalises BGP routes that keep being withdrawn and re-advertised (flapping). Each flap adds a penalty. When penalty exceeds threshold: route is suppressed (not used). Penalty decays over time. When penalty drops below reuse threshold: route is used again. Half-life: 15 minutes default (penalty halves every 15 min). Suppress limit: 2000. Configure: bgp dampening. Warning: aggressive dampening can delay legitimate route recovery. RIPE recommends against dampening for most routes today — modern BGP hardware can handle updates. Useful only for extremely unstable routes.

Q48 What is BGP aggregation?

Ans BGP aggregation combines multiple specific prefixes into one summary advertisement — reduces routing table size on internet. Two methods: aggregate-address 192.168.0.0 255.255.252.0 — advertises summary only if at least one component route exists in BGP table. aggregate-address 192.168.0.0 255.255.252.0 summary-only — suppresses specific routes, advertises only summary. aggregate-address ... as-set — includes AS-path info from all component routes (prevents information loss). aggregate-address ... attribute-map — set attributes on aggregate. Verify: show bgp ipv4 unicast (aggregate shows as 'a' flag).

Q49 How do you troubleshoot a BGP session that will not establish?

Ans BGP troubleshooting: show bgp summary — check peer state. Active = TCP connection failing (wrong IP, firewall blocking TCP 179, no route to peer). Idle = waiting (admin shutdown or error). OpenSent/OpenConfirm = negotiating (AS mismatch, authentication failure, capability mismatch). Established = working. Debug: debug ip bgp [IP] events (lightweight). Check firewall: show ip access-lists (TCP 179 blocked?). Check route to peer IP: show ip route [peer-IP]. Check authentication: neighbor [IP] password. Check update-source: neighbor [IP] update-source Loopback0.

Q50 What is BGP update-source and when is it needed?

Ans By default, a BGP router uses the interface closest to the peer as its source IP. For iBGP peers that use loopback addresses as neighbour IPs: the router must configure update-source to use its loopback as the source. `neighbor 10.0.0.2 update-source Loopback0`. Why? Loopback addresses are stable (never go down unless router fails). Physical interfaces can flap — if source interface goes down, BGP session drops even if another path to peer exists. For iBGP: always use loopback IPs for neighbours. For eBGP multihop: use update-source with `ebgp-multihop [TTL]`.

Q51 What is BGP route reflector cluster?

Ans For redundancy, deploy two Route Reflectors. To prevent routing loops between them: both RRs in same cluster get the same cluster ID: `bgp cluster-id 1.1.1.1`. When RR receives a reflected route with its cluster ID in the cluster-list: it knows the route has already been in this cluster and discards it — loop prevention. Two RRs in same cluster must be fully meshed with each other and with all clients (each client peers with both RRs). Client only needs to peer with the RRs — not with all other clients.

Q52 What is BGP for IPv6 (MP-BGP)?

Ans MP-BGP (Multi-Protocol BGP, RFC 4760) extends BGP to carry routing information for multiple address families beyond IPv4 unicast. Address families: IPv4 unicast (traditional), IPv4 multicast, IPv6 unicast, VPNv4/VPNv6 (MPLS L3VPN), L2VPN. Configure IPv6: `router bgp 65001` `address-family ipv6 unicast` `neighbor [IPv6-peer] activate`. BGP session itself can use IPv4 TCP (for iBGP using IPv4 addresses) while carrying IPv6 routes — or use IPv6 TCP. The same BGP peer can carry multiple address families simultaneously.

Q53 What is BGP conditional advertisement?

Ans Conditional advertisement sends a BGP route to a peer ONLY IF another specific route exists (or does not exist) in the BGP table. Use case: dual-homed customer wants to advertise specific routes to ISP-A only when the link to ISP-B is up (otherwise ISP-A route would attract traffic that has no return path). Configure: `neighbor [IP] advertise-map ROUTE-MAP non-exist-map EXIST-MAP`. `advertise-map`: what to advertise. `non-exist-map`: advertise only if this route does NOT exist in BGP table (or `exist-map`: only if it DOES exist).

Q54 What is BGP outbound route filtering (ORF)?

Ans ORF allows a BGP peer to send its inbound filter to its neighbour — the neighbour then applies that filter outbound, preventing the filtered routes from being sent in the first place. Benefit: reduces unnecessary BGP update traffic. Without ORF: peer receives all routes, applies filter locally — wastes bandwidth. With ORF: neighbour knows your filter and never sends filtered routes. Configure: neighbor [IP] capability orf prefix-list send (I will send my filter to you) + neighbor [IP] capability orf prefix-list receive (I will accept your filter and apply it outbound).

Q55 What is BGP maximum-prefix?

Ans maximum-prefix protects your router from being overwhelmed by too many routes from a peer. Configuration: neighbor [IP] maximum-prefix 1000 80 warning-only. 1000 = maximum prefixes allowed from this peer. 80 = generate warning at 80% of limit. warning-only = just warn, do not shut down session (omit for default: session shutdown when limit exceeded). Use case: protect against misconfigured peer advertising thousands of routes (could fill routing table and crash router). Essential for internet-connected routers. Verify: show bgp neighbors [IP] (see prefix count and limit).

Q56 What is BGP bestpath compare-routerid?

Ans The bgp bestpath compare-routerid command adds a deterministic tiebreaker at step 10 of BGP path selection. Without it: when two identical paths exist, the one that was learned first (oldest) is preferred — non-deterministic and changes after router reload. With compare-routerid: always prefer route from peer with lower Router ID — deterministic result. Configure: bgp bestpath compare-routerid. Best practice: always configure on route reflectors and when running multiple eBGP sessions to same neighbour (multi-path scenarios).

Q57 What is BGP allowas-in?

Ans BGP normally rejects routes with its own AS number in the AS-path — loop prevention mechanism. But in MPLS L3VPN hub-and-spoke topologies, spoke sites receive routes from hub that originated in the same customer AS — BGP blocks them. allowas-in permits this: neighbor [IP] allowas-in [count]. count = how many times own AS is allowed in path (default 1). Also useful in dual-homed scenarios where provider reflects your own routes back to you. Use carefully — disables an important loop prevention mechanism. Verify: show bgp ipv4 unicast neighbors [IP] (see allowas-in status).

Q58 What is BGP weight vs local-preference difference?

Ans Weight: Cisco proprietary, local to the router only (not sent to any peer). Range 0-65535 (higher=better). Default 0 for received routes. Use to prefer one peer over another on a single router. Local Preference: shared across entire AS via iBGP (not sent to eBGP peers). Range 0-4294967295 (higher=better). Default 100. Use to set a preference for all routers in your AS. Memory trick: Weight = works on single router. Local Preference = works across your whole AS. Weight is checked first in path selection — Weight overrides Local-Preference.

Q59 What is BGP synchronisation and why is it disabled?

Ans BGP synchronisation rule: do not use or advertise an iBGP-learned route unless the same route also exists in the IGP. Purpose: ensure all non-BGP routers in the AS can also forward the traffic (avoid routing black holes). Modern networks: no longer relevant because all routers in the AS either run BGP (full mesh or RR) or MPLS is used. Sync adds delay (wait for IGP) and causes problems. Disabled by default in IOS 12.2.8 and later. To verify: `show ip protocols | include synchronization`. If still enabled: no synchronization under router bgp.

Q60 What is BGP table version?

Ans The BGP table version increments every time the BGP table changes (new route, removed route, attribute change). You can track whether BGP is stable by monitoring the table version — if it keeps incrementing, routes are changing. `show bgp ipv4 unicast summary`: 'BGP table version is 1234'. If table version increases rapidly: route flapping somewhere. `show bgp ipv4 unicast neighbors [IP]`: 'Table version in the update group = 1234' and 'Neighbor table version' — these should be equal (neighbour is up to date). Mismatch means updates in progress.

SECTION 03

Layer 2 Technologies

Q61 – Q90

*Spanning Tree · EtherChannel · VLANs · RSTP · MST · Port Security · DHCP Snooping***Q61** What is RSTP and how does it improve on STP?

Ans RSTP (Rapid Spanning Tree Protocol, IEEE 802.1w) dramatically improves STP convergence — from 30-50 seconds to 1-2 seconds. Key improvements: New port roles: Alternate Port (instant standby for root port), Backup Port (standby for designated port). New states: Discarding (combines Blocking+Listening), Learning, Forwarding. Proposal/Agreement handshake: instead of waiting timers, switches negotiate directly — immediate port transition. Edge ports (PortFast): skip negotiation for end-device ports. RSTP is backward compatible with STP — runs both on same network.

TIP *RSTP is the CCNP default — understand every port role and state change!*

Q62 What is MST (Multiple Spanning Tree)?

Ans MST (IEEE 802.1s) maps multiple VLANs to a smaller number of STP instances — reducing overhead while keeping per-VLAN topology control. Problem with PVST+: 1000 VLANs = 1000 STP instances — very CPU intensive. MST solution: group VLANs into instances. Example: VLANs 1-100 run Instance 1, VLANs 101-200 run Instance 2. All VLANs in same instance share one STP topology. IST (Internal Spanning Tree) = Instance 0 — default instance that all other instances interact with. MST regions: switches with same region name, revision, and VLAN-to-instance mapping form a region.

Q63 What is PVST+ and Rapid PVST+?

Ans PVST+ (Per-VLAN Spanning Tree Plus) runs a separate STP instance per VLAN — allows different root bridges per VLAN for load balancing. Example: VLAN 10 root = Switch-A, VLAN 20 root = Switch-B — both uplinks actively carry traffic. Rapid PVST+ adds RSTP convergence to each per-VLAN instance. Default on Cisco switches. Disadvantage: scales poorly with many VLANs (100 VLANs = 100 STP instances). MST is the scalable alternative. Configure: spanning-tree mode rapid-pvst. Show: show spanning-tree vlan 10 (per-VLAN topology).

Q64 What is BPDU Guard and BPDU Filter?

Ans BPDU Guard: when a PortFast/edge port receives any BPDU (should not happen — PCs do not send BPDUs), the port is immediately shut down (err-disabled). Prevents rogue switches from connecting to user ports and disrupting STP. Configure globally: spanning-tree portfast bpduguard default. Or per port: spanning-tree bpduguard enable. BPDU Filter: stops port from sending or receiving BPDUs — completely disables STP on that port. Dangerous: if a loop exists, it will not be detected. Only use BPDU Filter on interfaces where STP is genuinely not needed (external links to non-STP devices).

Q65 What is STP root guard?

Ans Root Guard prevents a port from becoming a root port — if a superior BPDU is received on a root-guard-enabled port, the port enters root-inconsistent state (blocked). Use on ports connected to less trusted switches where you never want them to become root bridge. Example: distribution switch connected to access switches — access switches should never be root bridge. Configure: spanning-tree guard root on the port. When the superior BPDUs stop, port automatically recovers (unlike err-disable which requires manual recovery or errdisable recovery).

Q66 What is STP loop guard?

Ans Loop Guard prevents a non-designated port from transitioning to forwarding if it stops receiving BPDUs — which can happen with unidirectional link failures. Without loop guard: if BPDUs stop arriving, switch thinks it has become the designated port and transitions to forwarding — creating a loop. With loop guard: port enters loop-inconsistent state (blocked) when BPDUs stop. When BPDUs resume, port recovers automatically. Configure: spanning-tree loopguard default (globally on all non-designated ports) or per port: spanning-tree guard loop. Works on opposite end of the link from root guard.

Q67 What is UDLD (Unidirectional Link Detection)?

Ans UDLD detects unidirectional fibre links — where one fibre strand works (you can send) but the other is broken (you cannot receive, or vice versa). STP does not detect this — it assumes bidirectional if link is up. UDLD: sends periodic probes. If probe is not echoed back: link is unidirectional — port shut down. Two modes: Normal (disables port, generates syslog). Aggressive (retries 8 times before disabling — better for intermittent issues). Configure: udld enable or udld aggressive on interface. Critical for fibre uplinks between switches — unidirectional links cause STP loops.

Q68 What is EtherChannel load balancing?

Ans EtherChannel (LAG) distributes traffic across member links using a hashing algorithm. Traffic in the same flow always takes the same link — avoids out-of-order packets. Hashing methods: src-mac, dst-mac, src-dst-mac, src-ip, dst-ip, src-dst-ip, src-dst-port (L4 — most granular). Choose based on traffic pattern: if all traffic comes from one source MAC □ src-mac gives poor distribution □ use dst-ip or src-dst-port. Configure: port-channel load-balance src-dst-ip. Verify: show etherchannel load-balance, show etherchannel detail (see per-member traffic distribution).

Q69 What is LACP vs PAGP for EtherChannel?

Ans LACP (Link Aggregation Control Protocol, IEEE 802.3ad): open standard, supported by all vendors. LACP modes: active (initiates negotiation), passive (responds but does not initiate). Both active is fine; one active + one passive works; both passive = no EtherChannel. PAGP (Port Aggregation Protocol): Cisco proprietary. PAGP modes: desirable (initiates), auto (responds). Both desirable works; desirable + auto works; both auto = no EtherChannel. Static EtherChannel (mode on): no negotiation — both sides must be on. Prefer LACP in multi-vendor environments. Verify: show lacp neighbor, show pagp neighbor.

Q70 What is VTP (VLAN Trunking Protocol) version 3?

Ans VTPv3 improves on VTPv1/v2: extended VLAN range support (VLANs 1-4094), MST instance propagation, primary server concept (only one primary server can make changes — prevents accidental overwrites), propagates private VLANs. VTPv3 modes: Server (create/delete VLANs), Client (cannot create VLANs), Transparent (does not participate, passes VTP), Off (completely ignore VTP). Security: VTP password mandatory in v3. Best practice for most networks: use VTP transparent mode or VTPv3 with strong password — accidental VLAN deletion from VTP is a classic network outage cause.

Q71 What is private VLAN (PVLAN)?

Ans Private VLANs isolate ports within the same VLAN — useful for hosting environments where servers share a subnet but must not communicate directly. PVLAN structure: Primary VLAN (outer container). Secondary VLANs: Isolated (port can only talk to promiscuous port — not to other isolated ports), Community (ports can talk to each other and to promiscuous port). Promiscuous port (usually the router/gateway): can communicate with all secondary VLANs. Use case: hotel/ISP where each customer gets isolation but shares same IP subnet. Configure on Cisco: vlan 100 private-vlan primary, vlan 101 private-vlan isolated.

Q72 What is DHCP snooping and how does it work?

Ans DHCP snooping prevents rogue DHCP servers from assigning wrong IPs (leading to MITM attacks). Switch ports are marked trusted (connected to real DHCP server) or untrusted (all client ports). Untrusted ports: DHCP Offer and Acknowledge messages are dropped — only clients can send Discover and Request on untrusted ports. Trusted ports: DHCP server replies allowed. DHCP snooping binding table records: client MAC, IP assigned, VLAN, and port. This table is used by Dynamic ARP Inspection and IP Source Guard. Configure: `ip dhcp snooping vlan 10` `ip dhcp snooping trust` (on uplink).

Q73 What is Dynamic ARP Inspection (DAI)?

Ans DAI prevents ARP spoofing attacks using the DHCP snooping binding table. When an ARP packet is received on an untrusted port: DAI checks if source MAC + source IP matches binding table. If match: allow ARP. If no match: drop ARP (likely spoofed). Trusted ports: bypass DAI check. ARP ACLs: for statically-addressed hosts (not in DHCP binding table): create ARP access-list. Configure: `ip arp inspection vlan 10`. Verify: `show ip arp inspection statistics` (see dropped packets). Common mistake: forgetting to trust the uplink port — causes legitimate ARP to be dropped.

Q74 What is IP Source Guard?

Ans IP Source Guard uses the DHCP snooping binding table to filter traffic on untrusted ports. Only traffic matching the binding table (correct IP + MAC for that port) is forwarded. Any packet with a spoofed source IP is dropped. Without IP Source Guard: a host can manually set any IP and send traffic. With it: only the DHCP-assigned IP is allowed on that physical port. Configure: `ip verify source vlan dhcp-snooping` on the interface. Add port security to also check MAC: `ip verify source vlan dhcp-snooping port-security`. Essential for high-security campus environments.

Q75 What is 802.1X port authentication?

Ans 802.1X controls network access at switch port level. Three roles: Supplicant (client device with 802.1X software), Authenticator (switch), Authentication Server (RADIUS). Process: client plugs in → switch requests authentication → client sends credentials → switch forwards to RADIUS → RADIUS accepts/rejects → switch opens/closes port. Authentication result VLANs: Auth VLAN (full access), Unauthenticated VLAN (limited access for phones), Guest VLAN (no 802.1X client). Configure: `aaa new-model` → `dot1x system-auth-control` → interface: `dot1x port-control auto`.

TIP *802.1X is critical for zero-trust campus security – common CCNP scenario!*

Q76 What is storm control?

Ans Storm control limits broadcast, multicast, or unknown unicast traffic to protect the network from broadcast storms or DoS attacks. Configure threshold as percentage of bandwidth or in packets-per-second. When exceeded: drop or shutdown port. Configure: `storm-control broadcast level 20 10` (shut when 20% exceeded, restart when drops below 10%). `storm-control action shutdown` (err-disable port instead of just dropping). Verify: `show storm-control` (per-interface threshold and status). Complements STP — STP prevents loops but storm control protects against excessive traffic even without loops.

Q77 What is SPAN, RSPAN, and ERSPAN?

Ans SPAN (Switched Port Analyzer): mirrors traffic from monitored ports to a local destination port (for packet analyzer). Source = monitored ports, Destination = analyzer port. On same switch only. RSPAN (Remote SPAN): source and destination on different switches — uses a special RSPAN VLAN to carry mirrored traffic across trunk links. ERSPAN (Encapsulated Remote SPAN): carries mirrored traffic over routed IP networks (GRE encapsulation) — destination can be anywhere on the network. Used for: intrusion detection, traffic analysis, network forensics. Configure: `monitor session 1 source interface Gi0/1`, `monitor session 1 destination interface Gi0/2`.

Q78 What is VRF (Virtual Routing and Forwarding)?

Ans VRF creates multiple independent routing tables on one router — like having multiple virtual routers. Each VRF has its own routing table, interfaces, and routing protocols. Uses: MPLS L3VPN (customer isolation), overlapping IP address spaces, management separation. Configure: `ip vrf CUSTOMER-A` `rd 65001:1` `route-target export 65001:1` `route-target import 65001:1`. Assign interface: `ip vrf forwarding CUSTOMER-A` (on interface — this removes existing IP). Add routes: `ip route vrf CUSTOMER-A 0.0.0.0/0 [next-hop]`. Ping in VRF: `ping vrf CUSTOMER-A 192.168.1.1`.

Q79 What is Flex Links?

Ans Flex Links is a Cisco alternative to STP for active-standby redundancy on access switches. Two ports are configured as a Flex Link pair — one active, one standby. When active link fails: standby immediately becomes active (faster than STP reconvergence). Key advantage: predictable and simpler than STP. Used in simple topologies where you want guaranteed active-standby without STP complexity. Configure: `interface [active-port] switchport backup interface [standby-port]`. Verify: `show interfaces switchport backup detail`. Note: Flex Links does not support load balancing across both links simultaneously.

Q80 What is the difference between access VLAN and voice VLAN?

Ans Access VLAN: carries data traffic from PC or device connected to the port — untagged. Voice VLAN (also called auxiliary VLAN): carries VoIP traffic from IP phone — tagged with voice VLAN ID. An IP phone has a built-in mini-switch: data from PC passes through untagged (access VLAN), phone traffic is tagged with voice VLAN ID. Configure: `switchport access vlan 10` (data VLAN) + `switchport voice vlan 20` (voice VLAN). The switch automatically trusts CoS bits from the phone for QoS and places phone in voice VLAN. Verify: `show interfaces Gi0/1 trunk` (see both VLANs active).

Q81 What is STP portfast?

Ans PortFast is a Cisco feature allowing an access port to immediately transition to forwarding state, skipping the 15-second Listening and 15-second Learning states. Only use on ports connected to end devices (PCs, printers, servers) — never on switch-to-switch links (loops would form before STP could prevent them). Configure per port: `spanning-tree portfast`. Or globally on all access ports: `spanning-tree portfast default`. Best practice: always combine with BPDU Guard. Without BPDU Guard, if someone plugs a switch into a PortFast port, it could disrupt STP.

Q82 What is MAC address flapping?

Ans MAC flapping occurs when a MAC address is learned on one port then immediately on a different port — the switch keeps updating its MAC table. Causes: network loop (most common and serious — implies STP failure), legitimate host dual-homed (connected to two ports — fine with bonding), hardware issue. Symptom: show log shows 'MAC ADDR MOVE NOTIFICATION' repeatedly. Diagnose: show mac address-table count (is table growing rapidly?), show mac address-table address [MAC] (which ports?). If loop: check STP status immediately — may need to disconnect redundant links.

Q83 What is carrier delay on a switch interface?

Ans Carrier delay is the minimum time an interface must be in the up or down state before notifying the routing protocol. This prevents brief link bounces (milliseconds-long flaps) from triggering unnecessary routing reconvergence. Configure: carrier-delay up [milliseconds] [down milliseconds]. For uplinks: carrier-delay up 2000 (wait 2 seconds after link comes up before telling OSPF/EIGRP). Different from hold-time: carrier-delay is per-physical-interface physical state notification; hold-time is per-routing-protocol notification. Use when physical links are unstable.

Q84 What is the difference between trunk and routed port on a switch?

Ans Trunk port: Layer 2 switch port carrying multiple VLANs with 802.1Q tags. No IP address on physical interface. Used for: switch-to-switch uplinks, switch-to-router ROAS, hypervisor uplinks. Routed port: Layer 3 port with an IP address — acts like a router interface. Does not participate in spanning tree as a switch port. No VLANs. Used for: switch-to-router direct connections, Layer 3 switch-to-router uplinks. Configure: no switchport (removes Layer 2 capabilities), then ip address. Verify: show interfaces Gi0/1 switchport (shows 'Switchport: Disabled' for routed port).

Q85 What is STP timer manipulation?

Ans STP timers control convergence speed. Hello timer (2s default): how often Root Bridge sends BPDUs. Forward Delay (15s default): time in Listening and Learning states. Max Age (20s default): how long switch stores BPDU before discarding (declares root bridge lost). Tuning rules: only change timers on Root Bridge (others get timers from Root Bridge's BPDUs). Formula: $2 * (\text{forward-delay} - 1) \geq \text{max-age}$. $2 * (\text{max-age} - 1) \geq \text{hello-time}$. Violation causes topology instability. In modern networks: use RSTP instead of STP timer tuning — RSTP is faster and does not need timer adjustment.

Q86 What is STP topology change notification (TCN)?

Ans When a port transitions to forwarding or goes down, a Topology Change Notification (TCN) is sent to the Root Bridge. Root Bridge then sets the TC (Topology Change) flag in BPDUs for a period = max-age + forward-delay (35 seconds). All switches receiving TC BPDUs reduce their MAC address table aging time from 300 seconds to forward-delay (15 seconds). Effect: MAC table entries age out quickly and switches must relearn MAC addresses. Problem: in large networks, TCNs cause massive MAC flooding temporarily. RSTP: handles topology changes more intelligently — only flushes MACs on relevant ports.

Q87 What is the native VLAN security concern?

Ans Native VLAN traffic (VLAN 1 by default) traverses trunk links untagged. Security concern: VLAN hopping attack — attacker sends double-tagged frames (outer tag = native VLAN, inner tag = target VLAN). First switch strips outer tag (native VLAN untagged). Second switch sees inner tag and forwards to target VLAN — bypassing VLAN isolation. Mitigations: change native VLAN to an unused VLAN ID (switchport trunk native vlan 999), prune native VLAN from all trunks (switchport trunk allowed vlan remove 999), enable vlan dot1q tag native (tag native VLAN traffic).

Q88 What is Flex Link preemption?

Ans Flex Link preemption allows the primary link to automatically reclaim active status when it recovers after a failure. Without preemption: after primary link failure and recovery, the backup link remains active — traffic continues on backup. With preemption (configured with delay): primary link becomes active again after a configurable delay (to ensure stability). Configure: interface [active-port] switchport backup interface [standby-port] preemption mode forced delay 50. delay 50 = wait 50 seconds before preempting to ensure primary link is stable. Similar concept to HSRP preempt.

Q89 What is STP port priority?

Ans STP port priority determines which port becomes root port when multiple ports on a switch connect to the same upstream switch (equal-cost paths to root bridge). Lower port priority = preferred root port. Default: 128. Range: 0-240 (in multiples of 16). Configure: spanning-tree [vlan ID] port-priority [0-240] on the interface. The full port ID = priority + port number — lower port ID = preferred. Used for: controlling which physical port is used when you have redundant uplinks with equal-cost paths. Only affects the downstream switch's port selection, not the upstream switch.

Q90 What is MAC move tracking?**Ans**

MAC move tracking monitors when a MAC address moves from one port to another and generates alerts — useful for detecting network loops (MAC address would appear on multiple ports rapidly) or identifying hosts that reconnect to different ports (asset tracking). Configure: `mac address-table notification change`. Enable SNMP notification: `snmp-server enable traps mac-notification`. MAC moves in a loop scenario: MAC table shows same MAC on multiple ports alternating rapidly — clearing MAC table helps temporarily but loop must be resolved. Verify: `show mac address-table notification change`.

SECTION 04	Wireless Networking	Q91 – Q120
802.11 Standards · WLC Architecture · RF Design · Security · FlexConnect · Roaming		
Q91	What are the 802.11 Wi-Fi standards a CCNP engineer must know?	
Ans	Key standards: 802.11a (1999): 5 GHz, up to 54 Mbps, OFDM modulation. 802.11b (1999): 2.4 GHz, 11 Mbps, DSSS modulation. 802.11g (2003): 2.4 GHz, 54 Mbps, backward compatible with b. 802.11n (Wi-Fi 4, 2009): 2.4 + 5 GHz, 600 Mbps, MIMO. 802.11ac (Wi-Fi 5, 2013): 5 GHz only, 3.5 Gbps, MU-MIMO. 802.11ax (Wi-Fi 6, 2019): 2.4+5+6 GHz, 9.6 Gbps, OFDMA, BSS Colouring, TWT. 802.11be (Wi-Fi 7, 2024): up to 46 Gbps, multi-link operation. CCNP focuses on 802.11ac and 802.11ax.	
TIP	<i>Wi-Fi standards and 802.11ax (Wi-Fi 6) features are heavily tested in CCNP ENCOR!</i>	
Q92	What is OFDMA in Wi-Fi 6 and why does it help?	
Ans	OFDMA (Orthogonal Frequency Division Multiple Access) allows one AP transmission to serve multiple clients simultaneously by dividing a channel into smaller 'Resource Units (RUs)'. Traditional Wi-Fi (OFDM): one client per transmission — other clients wait. Wi-Fi 6 OFDMA: AP allocates different RUs to different clients in the same time slot — like a postal truck delivering to multiple houses in one trip instead of one at a time. Result: much better performance in dense environments (offices, hospitals, stadiums) where many clients transmit small packets simultaneously. Reduces latency dramatically.	
Q93	What is BSS Colouring in Wi-Fi 6?	
Ans	BSS Colouring assigns a 6-bit 'colour' number to each AP. When a client detects Wi-Fi signals from a different-coloured BSS (different network on same channel): it checks signal strength. If signal is below threshold (inter-BSS): transmit immediately instead of waiting (spatial reuse). Without colouring: all co-channel APs must wait for each other — wasting airtime. With colouring: overlapping APs on same channel can transmit simultaneously if their signals do not interfere. Dramatically increases channel reuse efficiency in dense deployments like enterprise campus floors.	

Q94 What is the Cisco wireless architecture (WLC + LWAP)?

Ans Cisco enterprise wireless uses Lightweight Access Point (LWAP) with a centralised Wireless LAN Controller (WLC). LWAPs are 'dumb' — they handle radio transmission but all intelligence (authentication, roaming, RF management, security) lives on the WLC. Communication: CAPWAP tunnels (UDP 5246 for control, 5247 for data) between LWAP and WLC. Cisco WLC models: 9800 series (hardware and virtual). Benefits: central management of thousands of APs, seamless roaming, consistent policy enforcement, automatic RF optimisation.

Q95 What is CAPWAP and how does it work?

Ans CAPWAP (Control and Provisioning of Wireless Access Points, RFC 5415) is the protocol between Lightweight APs and WLC. Two tunnels: Control channel (UDP 5246, DTLS-encrypted): AP registration, configuration, management commands. Data channel (UDP 5247, optionally encrypted): client data traffic from AP forwarded to WLC. AP discovery: LWAP broadcasts locally, then queries DNS for CISCO-CAPWAP-CONTROLLER, checks DHCP option 43, or uses configured IP. WLC selection: AP joins WLC with most capacity. Verify: show ap summary (see all APs and their WLC).

Q96 What is FlexConnect (H-REAP)?

Ans FlexConnect mode allows an AP at a remote site to locally switch client traffic — not sending all traffic back to WLC over WAN. Two sub-modes: Standalone mode (WAN down): AP uses cached config from WLC, locally authenticates clients and switches traffic. Connected mode (WAN up): control traffic goes to WLC, data locally switched. Critical for: branch offices on slow WAN links (avoid backhauling all user traffic to WLC at HQ), maintaining connectivity if WAN fails. Configure AP: ap [name] mode flex. Configure per-SSID: flexconnect local-switching.

Q97 What is the difference between local mode and FlexConnect mode?

Ans Local mode (centralized): ALL client data tunnelled back to WLC via CAPWAP. WLC switches traffic to correct VLAN. Advantage: central policy enforcement, easy management. Disadvantage: all traffic traverses WAN — wastes bandwidth for branch sites, WAN failure = clients lose internet. FlexConnect mode: client data switched locally at AP (stays on branch LAN). Advantage: WAN-efficient, survives WAN failure. Disadvantage: some features limited in standalone mode (certain security policies require WLC). Use local mode for campus APs near WLC; FlexConnect for remote branches.

Q98 What is RF band steering?

Ans Band steering encourages dual-band clients (supporting both 2.4 GHz and 5 GHz) to connect on 5 GHz. Why: 5 GHz is faster, less congested (fewer devices use it), more non-overlapping channels. Without band steering: clients often choose 2.4 GHz (stronger signal on older hardware). WLC band steering: delays 802.11 probe responses on 2.4 GHz for capable clients — client waits, does not hear 2.4 GHz response, connects to 5 GHz. Aggressive mode: actively disassociates clients from 2.4 GHz. Configure on WLC: WLAN □ Advanced □ Band Select. Verify: show advanced band-select.

Q99 What is RRM (Radio Resource Management) in Cisco WLC?

Ans RRM automatically manages RF channels and transmit power across all APs to optimise the wireless environment. Components: TPC (Transmit Power Control): reduces AP power when neighbours are too close — prevents co-channel interference. DCA (Dynamic Channel Assignment): automatically assigns channels to minimise interference. Coverage Hole Detection: increases AP power if clients report poor signal. Coverage Optimal: adjusts power to maintain minimum coverage. RRM runs globally by default. Configure: WLC □ Wireless □ 802.11a/b □ RRM. Manual override: ap channel and ap txpower commands. Verify: show advanced 802.11a channel.

Q100 What is wireless roaming and how does layer 2 vs layer 3 roaming work?

Ans Layer 2 roaming: client moves to a different AP in the same subnet (same VLAN). IP address unchanged. Fast and transparent — client keeps its IP. Layer 3 roaming: client moves to an AP on a different subnet. Client keeps original IP (mobility anchor maintains association). WLC tunnels traffic: anchor WLC (original) and foreign WLC (current) communicate — client traffic tunnelled from foreign back to anchor. This enables IP retention across subnets. For voice calls: must be transparent — IP change would drop call. Verify: show wireless client detail (see client mobility state).

Q101 What is 802.11r (Fast BSS Transition)?

Ans 802.11r Fast BSS Transition (FT) pre-authenticates a client with neighbouring APs before it roams — caches authentication keys at target AP. When client roams: immediate connection (no full authentication re-run). Critical for: VoIP (voice calls must not have >50ms interruption during roam). Without 802.11r: standard roam = 300ms+ (EAP authentication). With 802.11r: 50ms. Two modes: Over-the-Air FT (client communicates directly with target AP before roaming), Over-the-DS (client communicates with target AP via current AP). Configure on WLC SSID settings: Fast Transition = Adaptive.

Q102 What is 802.11k (Radio Resource Management)?

Ans 802.11k helps clients make better roaming decisions by providing a Neighbour Report — a list of nearby APs they could roam to. Without 802.11k: client must scan all channels to find APs (takes seconds). With 802.11k: AP sends client the pre-built neighbour list. Client scans only listed channels — much faster. Works together with 802.11v and 802.11r: k=find neighbours, v=BSS transition management, r=fast authentication. The combination is called 802.11kvr — enable all three for best enterprise roaming performance.

Q103 What is 802.11v (BSS Transition Management)?

Ans 802.11v allows the AP/WLC to suggest or force a client to roam to a better AP. Without 802.11v: client decides when to roam — may stay on a weak AP even when better ones are available (sticky client). With 802.11v: WLC detects poor RSSI, sends BSS Transition Management Request to client suggesting better AP. Client can accept or reject. Disassociation imminent bit: if AP needs to disable for maintenance, it warns clients to roam before going down. Enables load balancing across APs (WLC can direct clients away from overloaded AP).

Q104 What is SSID and how many SSIDs per AP is recommended?

Ans SSID (Service Set Identifier) is the network name clients see. Each SSID on an AP creates additional overhead — beacons, probe responses, management traffic. Rule of thumb: maximum 4 SSIDs per radio band per AP. Beyond 4: significant overhead reduces client data throughput. Common SSID set: Corporate (WPA3-Enterprise, 802.1X), Voice (separate QoS settings, 802.11r), Guest (isolated, internet-only), IoT (isolated VLAN, limited access). More SSIDs = more overhead. Use fewer, purpose-designed SSIDs for best performance.

Q105 What is Wi-Fi signal strength and what is acceptable RSSI?

Ans RSSI (Received Signal Strength Indicator) is measured in dBm (negative values — closer to 0 = stronger). Guide: -30 dBm: Excellent (very close to AP). -50 to -60 dBm: Good (normal connected client). -70 dBm: Minimum for reliable data. -75 dBm: Minimum for VoIP (poor voice quality below this). -80 dBm: Minimum for basic connectivity. -90 dBm: Unusable. Enterprise design target: -65 to -67 dBm minimum at edge of coverage area. SNR (Signal to Noise Ratio): difference between signal and noise floor. SNR > 25 dB is good for data; > 35 dB for VoIP.

Q106 What is wireless QoS (WMM and QoS policies)?

Ans WMM (Wi-Fi Multimedia, IEEE 802.11e) provides four access categories: Voice (VO) — highest priority, Video (VI), Best Effort (BE), Background (BK). WMM maps Cisco QoS DSCP markings to WMM categories. VoIP packets (DSCP EF = 46) map to Voice category — lowest contention window (transmit first). Required for any enterprise network with voice or video. Configure on WLC SSID: QoS Profile = Platinum (voice), Gold (video), Silver (best effort), Bronze (background). APs enforce QoS when transmitting to clients.

Q107 What is the Cisco WLC 9800 architecture?

Ans Cisco Catalyst 9800 WLC is the current-generation wireless controller. Available as: Hardware appliance (9800-40, 9800-80), Virtual appliance (9800-CL — runs on ESXi, KVM, AWS, Azure), Embedded in Catalyst 9000 switches (9800-SW), Embedded in access points (9800-AP for small deployments). Key features: IOS XE based (not AireOS — same OS as Catalyst switches), hitless software upgrades, high availability (1:1 redundancy), scale up to 6000 APs per controller. Commands: show wireless ap summary, show wireless client summary.

Q108 What is AP Join Process in Cisco wireless?

Ans AP join process: AP powers on □ gets IP from DHCP. AP discovers WLC: DNS lookup (CISCO-CAPWAP-CONTROLLER), DHCP option 43 (WLC IP in hex), local subnet broadcast, manually configured (ap-manager IP). AP sends DTLS (CAPWAP) discovery request to WLC. WLC responds. AP downloads config and image from WLC (if software mismatch). AP forms CAPWAP control channel. AP transmits on assigned channels. Verify: show ap summary (see joined APs). If AP not joining: check DHCP option 43, check firewall allowing UDP 5246, check WLC certificate.

Q109 What is wireless site survey and its purpose?

Ans A wireless site survey determines optimal AP placement before deployment. Types: Predictive (pre-deployment): software simulation of RF coverage based on building floor plan. Active (post-deployment): walk with laptop measuring actual signal, throughput, and roaming behaviour. Factors: building materials (concrete walls heavily attenuate), interference sources (microwaves, other Wi-Fi), client density per area. Deliverables: heat map of coverage, channel/power plan, AP placement recommendations. Without site survey: dead zones, overlapping channels causing interference, poor roaming. Essential before any enterprise wireless deployment.

Q110 What is wireless security – WPA2/WPA3 enterprise?

Ans WPA2-Enterprise uses 802.1X – each user has unique credentials (no shared password). Flow: client connects → switch forwards to RADIUS (ISE or FreeRADIUS) → RADIUS verifies username/password against Active Directory → access granted (specific VLAN assigned based on user group). EAP types: EAP-TLS (certificate-based – most secure), PEAP-MSCHAPv2 (password + server cert – most common), EAP-FAST (Cisco proprietary). WPA3-Enterprise adds 192-bit security mode. Key advantage over PSK: revoke a specific user's access by disabling their AD account – no password change needed for all users.

Q111 What is Cisco ISE for wireless?

Ans Cisco Identity Services Engine (ISE) is a network access control platform. For wireless: acts as RADIUS server, authenticates clients via 802.1X, applies dynamic VLAN assignment and ACLs based on user identity. Profiling: identifies device type (iPhone, laptop, Android) and applies appropriate policy. Guest portal: web-based guest access with self-registration or sponsor approval. Posture assessment: checks if device has antivirus, OS updates before granting full access – non-compliant devices quarantined. CCNP requires understanding ISE integration with WLC, not deep ISE configuration.

Q112 What is wireless mesh networking?

Ans Wireless mesh allows APs to wirelessly backhaul traffic to other APs – no cable required to every AP. Some APs are Root APs (connected to wired network), others are Mesh APs (connect wirelessly to Root or other Mesh APs). Use cases: warehouses where cabling is difficult, outdoor campuses, construction sites. Performance consideration: each wireless hop degrades throughput (half available bandwidth for backhaul). Best practice: keep mesh depth to 1-2 hops. Configure in WLC: AP mode = Bridge or FlexConnect. Mesh uses 802.11 5GHz band for backhaul while serving clients on another band.

Q113 What is AP high availability in Cisco wireless?

Ans Two types: N+1 HA (network level): multiple WLCs — if one fails, APs join another WLC. Configure secondary WLC: each AP has primary, secondary, tertiary WLC configured. AP SSO (Stateful Switchover): 1:1 WLC pairs — standby WLC maintains complete copy of active WLC state. When active fails: standby takes over in milliseconds, APs keep all client sessions. AP does not need to re-join or re-authenticate clients. Requires same hardware model. Configure: redundancy-config (on both WLCs). Verify: show redundancy states (on WLC).

Q114 What is wireless location services?

Ans Wireless location services track client devices using signal strength (RSSI) triangulation from multiple APs. Cisco MSE (Mobility Services Engine) or CMX (Connected Mobile Experiences): collects RSSI readings from multiple APs, calculates client location (within 3-5 meters accuracy with proper AP density). Uses: asset tracking (medical equipment in hospital), customer analytics (retail foot traffic), security (locating devices that should not be on network). CCNP: understand how location works conceptually — not deep MSE configuration.

Q115 What is per-client rate limiting in wireless?

Ans Per-client rate limiting prevents one wireless client from consuming all available bandwidth (especially important for guest networks). Configure on WLC: WLAN ▢ QoS ▢ Bandwidth contracts: Average Data Rate (long-term average), Burst Data Rate (short-term peak). Example: 10 Mbps average, 20 Mbps burst for guest clients. Configure per-SSID or per-user (with ISE sending rate limit attributes in RADIUS). Benefits: fair bandwidth sharing, prevents video streaming from impacting business traffic on guest SSID. Per-AP rate limits also available.

Q116 What is wireless multicast optimisation?

Ans Wireless multicast is inefficient by default — multicast traffic sent at lowest mandatory rate (1 Mbps in 2.4 GHz), using most of airtime. Solutions: IGMP Snooping on WLC: only send multicast to APs with interested clients (not all APs). Multicast-Unicast conversion: WLC converts multicast to unicast for small groups (better performance). Optimised Multicast Video: WLC streams video multicast at a controlled rate. For large video deployments: use DirectStream feature. Configure: WLAN ▢ Advanced ▢ Multicast/Unicast. Critical for wireless video surveillance and IPTV deployments.

Q117 What is the difference between 2.4 GHz channel planning and 5 GHz?

Ans 2.4 GHz: Only 3 non-overlapping 20 MHz channels in most countries (1, 6, 11). Severe co-channel interference in dense deployments. Adjacent APs should use different channels in a honeycomb pattern: 1-6-11-1-6-11. Limited to 3 cells per floor before reuse. 5 GHz: 25 non-overlapping 20 MHz channels available (varies by country). Use 40 MHz or 80 MHz channels in 5 GHz for more throughput. 5 GHz signal attenuates faster — requires more APs for same coverage area. DCA (Dynamic Channel Assignment) on WLC automates channel selection.

Q118 What is RRM transmit power control?

Ans TPC (Transmit Power Control) automatically adjusts AP power to maintain optimal cell size. Too much power: APs overlap heavily — clients see many APs at equal strength — causes stickiness (client does not roam to better AP). Too little power: coverage holes. RRM TPC algorithm: each AP checks RSSI of neighbouring APs. Target: -70 dBm from neighbours. If neighbours are stronger: reduce power. If weaker: increase power. Configure threshold: on WLC. Manual override: `ap txpower [ap-name] 802.11a/b [power-level]`. Verify: `show advanced 802.11a txpower`.

Q119 What is Cisco CleanAir technology?

Ans CleanAir is Cisco's spectrum intelligence technology built into Wi-Fi 6 APs. It detects, classifies, and locates non-Wi-Fi interference sources: microwave ovens, Bluetooth devices, ZigBee, baby monitors, frequency jammers, and more. APs use dedicated spectrum analysers. When CleanAir detects interference: it classifies the source, assigns a severity score, and recommends channel change. DCA incorporates CleanAir data to automatically avoid interference channels. View interference: WLC □ Monitor □ Spectrum Expert □ Air Quality Report. Very useful in dense enterprise environments.

Q120 What is wireless network management with Cisco DNA Center?

Ans Cisco DNA Center manages the entire network including wireless — switches, routers, APs, and WLCs — from a single dashboard. Wireless-specific features: Assurance (AI-powered wireless health monitoring, client experience analytics), automated wireless provisioning (deploy new APs with pre-configured settings), network hierarchy (sites/buildings/floors with RF floor plans), software upgrades (schedule across all APs), and rogue AP detection. CCNP requires understanding of DNA Center architecture and its role in intent-based networking, not deep DNA Center administration.

SECTION 05

SD-WAN & SD-Access

Q121 –
Q150

Cisco SD-WAN · vManage · SD-Access · SDA Fabric · DNA Center · Zero Trust

Q121 What is Cisco SD-WAN and why are enterprises adopting it?

Ans Cisco SD-WAN (formerly Viptela, now integrated into IOS XE) is a software-defined WAN solution that separates network control from forwarding and centralises management. Why enterprises adopt it: Replaces expensive MPLS with cheaper broadband + LTE (50-80% WAN cost reduction). Application-aware routing: automatically routes VoIP over best-performing link, other traffic over cheaper links. Zero-touch deployment: new branches provisioned in hours without sending engineers. Central policy management: one policy change applies to all 500 branches instantly. Cloud optimisation: direct cloud breakout (Office 365, AWS) instead of backhauling to data centre.

TIP SD-WAN is ~10% of CCNP ENCOR exam. Understand architecture and components!

Q122 What are the SD-WAN components?

Ans vManage: centralised management and policy plane (web GUI and REST API). All configuration changes made here. vSmart controller: centralised control plane — distributes routing and policy information to all WAN edges. vBond orchestrator: initial device onboarding and authentication — first point of contact for new devices, provides vManage and vSmart addresses. WAN Edge router (vEdge or IOS XE SD-WAN router): data plane device at each site — forwards traffic based on policies from vSmart. OMP (Overlay Management Protocol): routing protocol between SD-WAN components — carries routes and policies.

Q123 What is the SD-WAN overlay and underlay?

Ans Underlay: physical transport networks — MPLS, broadband internet, 4G/5G LTE, metro Ethernet. SD-WAN does not replace these — it overlays them. Overlay: IPsec tunnels built on top of underlay transport. WAN edge routers form IPsec tunnels to other edges through vSmart. Traffic flows through overlay tunnels — encrypted and transport-independent. Benefits: if MPLS link fails, traffic automatically fails over to broadband or LTE overlay within seconds. Application policies route traffic across the best underlay for each application.

Q124 What is OMP (Overlay Management Protocol)?

Ans OMP is SD-WAN's control plane protocol — like BGP for the SD-WAN overlay. Runs between: WAN edges and vSmart controllers. Carries: IPv4/IPv6 routes, service routes (which services each site offers), TLOCs (Transport Locators — where to build tunnels to reach each site). vSmart acts like a route reflector — receives routes from all edges, applies policies, and distributes to other edges. WAN edges only form OMP with vSmart (not directly with each other) — centralised control. OMP runs over DTLS/TLS (encrypted).

Q125 What is a TLOC (Transport Locator) in SD-WAN?

Ans TLOC identifies a specific WAN transport interface on a WAN edge router. Format: System-IP + Colour (transport label) + Encapsulation (IPsec or GRE). Example: System-IP=10.0.0.1, Colour=mpls, Encapsulation=ipsec and System-IP=10.0.0.1, Colour=biz-internet, Encapsulation=ipsec. One site might have 3 TLOCs: one MPLS, one broadband, one LTE. vSmart distributes TLOC information — other edges use TLOCs to build IPsec tunnels. Application policies specify which TLOCs to use for different traffic types.

Q126 What is SD-WAN application-aware routing?

Ans Application-aware routing uses SLA (Service Level Agreement) probes to continuously measure latency, jitter, and packet loss on each WAN transport. Each application class has SLA requirements: VoIP needs <150ms latency, <30ms jitter, <1% loss. If a transport link's measured performance degrades below application SLA threshold: SD-WAN automatically moves that application's traffic to a better-performing transport — in real time. This is impossible with traditional WAN where routing is topology-based, not performance-based.

Q127 What is SD-WAN zero-touch provisioning (ZTP)?

Ans ZTP allows a new WAN edge router to automatically join the SD-WAN overlay with no manual configuration at the branch. Process: ship box to branch, branch plugs it in. Device contacts Cisco ZTP portal (cloud) or on-premises vBond. vBond authenticates the device (using pre-installed certificate), provides vManage and vSmart addresses. Device downloads configuration template from vManage. Full SD-WAN functionality in 30-60 minutes — no engineer visit needed. Requires: serial number pre-registered in vManage, configuration template pre-built, WAN connectivity at branch.

Q128 What is SD-WAN data policy vs application policy?

Ans Two policy types in SD-WAN vManage: Data policy (applied centrally via vSmart, executed on WAN edge): controls what to do with traffic matching criteria (accept, drop, redirect to specific TLOC, count). Examples: block P2P, redirect social media to direct internet breakout, force VoIP over MPLS. Application policy: configured on vManage, defines which applications map to which SLA class and which TLOCs to prefer. Data policies are stateless; some are stateful (firewall features). Policies distributed from vSmart to all edges — change once, applies everywhere.

Q129 What is SD-Access and how does it differ from traditional campus?

Ans SD-Access (Software-Defined Access) is Cisco's intent-based networking solution for campus networks. Traditional campus: manually configure VLANs, ACLs, STP, and port security on each switch separately. SD-Access: define policy once in DNA Center ('HR group can access HR servers'), DNA Center automatically translates to configuration across all switches. Uses VXLAN+LISP overlay — scales beyond 4094 VLAN limit, enables seamless Layer 2/3 mobility. Zero Trust: security is user-and-device-identity-based, not VLAN-based. Auto-provisions new switches.

Q130 What is the SDA fabric?

Ans SD-Access fabric is the overlay network providing any-to-any connectivity with consistent policy. Underlay: physical campus switches/routers running IS-IS (auto-configured by DNA Center). Overlay: VXLAN tunnels carrying traffic between fabric nodes. Three roles: Edge Node (access layer — connects endpoints). Border Node (connects fabric to external networks — WAN, data centre, internet). Control Plane Node (LISP map server — tracks where endpoints are). Fabric enables: micro-segmentation (endpoints in same VLAN but different policies), scalable mobility, and consistent policy across all campus locations.

Q131 What is LISP in SD-Access?

Ans LISP (Locator/ID Separation Protocol) separates device identity (EID — endpoint IP) from location (RLOC — fabric node IP). In SDA: Control Plane Nodes are LISP map servers — they know which EID is behind which Edge Node (RLOC). When traffic arrives for unknown destination: Edge Node queries Control Plane Node 'where is 192.168.1.100?' □ responds with fabric Edge Node's RLOC □ Edge Node builds VXLAN tunnel to that RLOC and forwards traffic. This enables scalable, IP-mobility-agnostic forwarding — clients move between Edge Nodes without routing protocol changes.

Q132 What is VXLAN in SD-Access?

Ans VXLAN (Virtual Extensible LAN) creates the overlay tunnels in SD-Access fabric. It encapsulates Layer 2 Ethernet frames in UDP packets (port 4789) for transport over the Layer 3 underlay. VNI (VXLAN Network Identifier): 24-bit identifier (16 million possible segments vs 4094 VLANs) — represents a virtual network segment. Each SGT (Scalable Group Tag) maps to a VNI. VXLAN enables: extending Layer 2 domains over Layer 3 boundaries, micro-segmentation using different VNIs per security group, seamless workload mobility. Fabric nodes act as VTEPs (VXLAN Tunnel Endpoints).

Q133 What is Scalable Group Tag (SGT) in SD-Access?

Ans SGT is a 16-bit tag assigned to a user or device based on identity (who you are) — not based on IP address or VLAN. Policy: 'Marketing group (SGT 10) cannot access Finance servers (SGT 20)' — enforced everywhere in fabric regardless of where Marketing users connect. SGTs follow the user across the entire campus. Security Group ACLs (SGACLs): matrix defining which SGT groups can communicate. Benefit: consistent policy for mobile users — plug in at any building, policy follows. Programmed by DNA Center/ISE, enforced by fabric Edge Nodes.

Q134 What is DNA Center in SD-Access?

Ans Cisco DNA (Digital Network Architecture) Center is the intent-based networking management platform. For SD-Access: Design (network hierarchy — sites, floors), Provision (automatic device configuration, add new switch in minutes), Policy (define who can access what — translated to SGTs and SGACLs), Assurance (AI/ML-driven issue detection, client experience tracking). DNA Center talks to network devices via NETCONF/YANG, REST API, and CLI. Replaces manual configuration across hundreds of switches with centralised intent-based policy.

Q135 What is SD-WAN Direct Internet Access (DIA)?

Ans Direct Internet Access (DIA) breaks internet-bound traffic out directly at the branch — bypassing the data centre. Traditional model: all branch internet traffic tunnelled to data centre □ hairpinned to internet (adds 50-100ms latency for cloud apps). DIA: branch router sends Office 365, Google Workspace, SaaS traffic directly to internet from branch (much lower latency). SD-WAN makes this safe: SD-WAN firewall inspects outbound traffic at branch, split tunnelling policy routes corporate traffic to MPLS/tunnel and cloud traffic to DIA. Critical for cloud-first enterprise strategy.

Q136 What is SD-WAN Cloud OnRamp?

Ans Cloud OnRamp optimises SD-WAN connectivity to cloud services: Cloud OnRamp for SaaS (Office 365, Salesforce): measures performance to SaaS applications from different branches through different paths — automatically routes each branch's SaaS traffic through the best gateway. Cloud OnRamp for IaaS (AWS, Azure): deploys virtual WAN edge routers in cloud regions — enterprise extends SD-WAN overlay into cloud, same policies apply. Cloud OnRamp for Colocation: extends to colocation facilities. Eliminates need for MPLS into data centre for cloud access.

Q137 What is Cisco SD-WAN security features?

Ans SD-WAN includes security directly in WAN edge: Zone-Based Firewall (ZBFW): stateful firewall — define zones (LAN, WAN, internet) and policies between them. DNS-layer security (Cisco Umbrella integration): blocks malicious domains at DNS level. Intrusion Prevention System (IPS): snort-based IPS running on WAN edge. SSL Proxy: decrypt and inspect HTTPS traffic. URL Filtering: block categories of websites. Cloud Security (Zscaler integration): redirect internet traffic to Zscaler for cloud-based inspection. All configured via vManage — consistent policy across all branches.

Q138 What is the SD-Access border node?

Ans Border Node connects the SD-Access fabric to external networks. Types: Default Border (handles unknown routes — sends to external network/data centre). Exit Border (connects to WAN, data centre, internet — specific routes). External Border (peers with traditional network using BGP or OSPF). Traffic leaving fabric: Edge node queries LISP Map Server □ if destination unknown □ traffic sent to Border Node □ Border Node routes via external network. Return traffic: enters Border Node □ LISP lookup to find destination Edge Node □ VXLAN tunnel to Edge. Separating roles: one border per WAN link is cleanest design.

Q139 What is DNA Center Assurance?

Ans DNA Center Assurance provides AI/ML-powered network health monitoring and troubleshooting. Features: Client 360 (complete history of any client — wired/wireless, authentication, DHCP, DNS, application performance). Device 360 (health of any device — CPU, memory, interface errors, neighbour changes). Issue detection (AI compares against baseline — 'client VLAN mismatch' automatically detected and root cause identified). Network time travel (look back at past state — 'what was the state 2 hours ago when the complaint happened?'). Dramatically reduces MTTR (Mean Time to Resolve) compared to manual CLI troubleshooting.

Q140 What is SDA Macro-segmentation vs Micro-segmentation?

Ans Macro-segmentation (traditional VLANs): coarse separation — all of HR in VLAN 10, all Finance in VLAN 20. 4094 VLAN limit. Policy based on IP ranges. Micro-segmentation (SDA with SGTs): granular — HR staff (SGT 10) vs HR contractors (SGT 11) vs HR servers (SGT 12) — even if they share same IP subnet. 65534 SGT values available. Policy in SGACL matrix: HR staff can access HR servers, HR contractors cannot. Zero Trust: every communication evaluated against SGT policy, not just VLAN membership. Dramatically reduces attack blast radius — compromise of one SGT cannot automatically access others.

Q141 What is the SD-WAN vManage NMS?

Ans vManage is the SD-WAN network management system — single pane of glass for the entire SD-WAN deployment. Functions: Centralised configuration (templates applied to all edge routers). Real-time monitoring (all tunnels, application performance, SLA compliance). Policy management (create and push data policies, security policies). Software management (schedule IOS XE upgrades across all edges). Alarms and events (site down, tunnel down, SLA violation). REST API: all vManage functions accessible via API for automation. CLI access still available on edge routers for direct access.

Q142 What is SDA policy plane?

Ans SDA policy plane defines who can communicate with whom based on identity. DNA Center provides GUI to: define user groups and device groups, map groups to SGTs, build SGACL policy matrix (allow/deny between SGT pairs). ISE assigns SGTs during authentication — user logs in □ ISE sees they are in HR AD group □ assigns SGT 10 □ fabric enforces HR policy. Policy is distributed to all fabric nodes. Enforcement points: Edge Nodes (inline traffic inspection), Border Nodes (policy enforced on inter-fabric traffic). Changes: update policy in DNA Center □ automatically pushed to all enforcement points.

Q143 What is the difference between traditional WAN and SD-WAN?

Ans Traditional WAN: MPLS circuits leased from provider, fixed bandwidth, expensive, long provisioning times (weeks). Configuration change: log into each router individually. No application visibility — just routing. SD-WAN: use any transport (MPLS + internet + LTE), abstracted by software overlay. Configuration: push from vManage to all sites. Application-aware routing: VoIP gets best path automatically. Cost: 40-70% reduction. Provisioning: new site in hours (ZTP). Centralised visibility: all application and performance data in vManage dashboard. DIA: local internet breakout improves cloud app performance.

Q144 What is OMP route advertisement in SD-WAN?

Ans WAN edge routers advertise their local routes to vSmart via OMP. Routes include: site prefixes (LAN networks behind each edge), service routes (firewall, IPS — which edges offer services). vSmart aggregates all routes, applies control policies, and distributes relevant routes to other edges. Edge routers install OMP routes in routing table — traffic forwarded to correct site through VXLAN/IPsec overlay. OMP vs BGP: OMP is simpler, designed specifically for SD-WAN. Key difference: OMP routes include TLOC information (which transport to use) — BGP routes do not have this concept.

Q145 What is Cisco DNAC vs vManage?

Ans DNA Center (DNAC): manages campus and branch networks — switches, routers, wireless controllers, APs. Uses SD-Access for campus automation. Provides assurance and analytics for campus/WAN/wireless. vManage: manages SD-WAN overlay specifically — WAN edge routers and their overlay connectivity. Integration: DNA Center integrates with vManage — see SD-WAN health in DNA Center dashboard alongside campus network. Together they provide end-to-end visibility from user device (campus) through WAN (SD-WAN) to application. CCNP requires understanding both architectures.

Q146 What is SDA Anycast Gateway?

Ans Anycast Gateway gives every Edge Node in the SDA fabric the SAME IP address as the default gateway for a given subnet — without FHRP (HSRP/VRRP). How: every Edge Node is configured with the same gateway IP and same gateway MAC. Client ARPs for gateway — any Edge Node responds with the same gateway MAC. When client roams to different Edge Node: gateway IP/MAC unchanged — no need for ARP update or IP change. This enables seamless Layer 2 mobility at Layer 3 scale. Configure: ip address [gateway] [mask] fabricpath on SVI.

Q147 What is SD-WAN multi-cloud?

Ans SD-WAN multi-cloud extends the SD-WAN fabric to multiple public clouds simultaneously — AWS, Azure, GCP. Virtual WAN edge routers (cloud-native vEdge/CSR1000v/Catalyst 8000V) deployed in cloud regions. Enterprise users connect to cloud resources via SD-WAN overlay — same policies apply as corporate WAN. Benefits: secure encrypted connectivity to cloud, centralised visibility (cloud traffic visible in vManage), consistent QoS and security policies in cloud. Cost comparison: SD-WAN cloud connectivity often cheaper and better-performing than Direct Connect/ExpressRoute for non-latency-critical workloads.

Q148 What is SD-Access underlay?

Ans SDA underlay is the physical IP network that the fabric overlay runs on. DNA Center auto-provisions the underlay: automatically configures IS-IS routing (no OSPF/BGP complexity), loopback IPs for all fabric nodes, OSPF-TE for traffic engineering (optional), PIM for multicast (optional). The underlay only needs to provide IP reachability between fabric nodes — VXLAN overlay tunnels carry everything else. Cisco recommends routed access design for SDA underlay (every access layer switch port is a /30 or /31 routed link — no STP needed). This eliminates STP complexity from campus design.

Q149 What is Cisco SD-WAN high availability?

Ans SD-WAN HA layers: Controller HA: vManage cluster (3 nodes for production), vSmart cluster (multiple vSmarts — edges connect to all). If one vSmart fails: edges still receive route and policy updates from remaining vSmarts. Data plane HA: each site has multiple WAN transports. If primary MPLS fails: traffic fails over to internet or LTE overlay in sub-second (BFD detects failure). Site redundancy: dual WAN edge routers at critical sites. Configuration: both edges share same site configuration. Failover: if active edge fails, standby takes over VXLAN tunnels.

Q150 What is SD-Access guest access?

Ans SD-Access guest access provides internet-only access to visitors without touching the corporate fabric. Approach: dedicated guest SSID □ maps to guest SGT □ SGACL denies guest from accessing corporate SGTs □ guest traffic directed to internet via Border Node DIA. Guest authentication options: Guest portal (ISE hosted web page — visitor registers, gets time-limited access), Sponsor portal (employee approves guest access), Social login (authenticate with Google/Facebook). Guest credentials valid for defined period (1 day, 1 week). No VPN, no corporate resource access — complete isolation enforced by SGACLs.

SECTION 06

Network Services

Q151 –
Q180

QoS · HSRP/VRRP/GLBP · NAT · IPv6 · DHCP · Multicast · NTP · Syslog

Q151 What is QoS and why is it needed?

Ans Quality of Service (QoS) prioritises certain types of traffic to ensure critical applications work well even when the network is congested. Without QoS: all traffic equal — a large file download can degrade a VoIP call. With QoS: VoIP packets get out first (strict priority queue), video gets guaranteed bandwidth, downloads use leftover capacity. QoS tools: Classification (identify traffic), Marking (tag with DSCP/CoS), Queuing (priority queues), Scheduling (how much bandwidth per queue), Policing (limit excess traffic), Shaping (smooth traffic bursts). Most important: mark traffic as close to source as possible.

TIP

QoS is ~10% of CCNP ENCOR. Know DSCP values and MQC configuration!

Q152 What are DSCP markings and their names?

Ans DSCP (Differentiated Services Code Point) is 6 bits in IP header. Key values: EF (Expedited Forwarding, DSCP 46): VoIP/real-time traffic — strict priority. CS5 (40): VoIP signalling. AF41 (34): interactive video (Webex, Teams). AF31 (26): critical data. CS3 (24): call signalling. AF21 (18): standard business data. CS1 (8): bulk/scavenger traffic. CS0 = 0: best effort (default). CoS (Class of Service): 3-bit field in 802.1Q VLAN tag — Layer 2 equivalent of DSCP. Mapping: DSCP EF maps to CoS 5, AF41 to CoS 4, etc.

Q153 What is MQC (Modular QoS CLI)?

Ans MQC is Cisco's QoS configuration framework. Three components: Class-map (match traffic): class-map match-any VOICE □ match dscp ef □ match dscp cs5. Policy-map (define actions per class): policy-map ENTERPRISE-QoS □ class VOICE □ priority percent 30 □ class VIDEO □ bandwidth percent 30 □ class class-default □ fair-queue. Service-policy (apply to interface): interface Gi0/0 □ service-policy output ENTERPRISE-QoS. The separation of matching, policy, and application makes QoS configurations modular and reusable.

Q154 What is LLQ (Low Latency Queuing)?

Ans LLQ (priority queue) gives VoIP traffic a strict priority — it is ALWAYS served before any other traffic. Configuration: class VOICE $\square$ priority percent 30 (VoIP gets up to 30% of bandwidth, always served first). If VoIP traffic is less than 30%: other queues get unused bandwidth. Policing: if VoIP exceeds 30% (misconfigured devices, attack): excess is dropped (not delayed). Why drop not delay? Delay is worse than loss for VoIP — delayed packets cause jitter. Requirements for good voice: latency <150ms end-to-end, jitter <30ms, packet loss <1%.

Q155 What is CBWFQ (Class-Based Weighted Fair Queuing)?

Ans CBWFQ provides guaranteed minimum bandwidth to traffic classes. Configuration: class VIDEO $\square$ bandwidth percent 25 (video guaranteed 25% of interface bandwidth). If video is using less than 25%: other classes get the excess. If interface is congested: video always gets its 25%. CBWFQ uses weighted fair queuing within each class — does not provide the strict priority LLQ provides. Typical policy: LLQ for VoIP (strict priority), CBWFQ for video and critical data (guaranteed bandwidth), fair-queue for class-default (remaining traffic).

Q156 What is traffic policing vs traffic shaping?

Ans Policing: enforces bandwidth limit by DROPPING excess traffic (above defined rate). Hard limit — no buffering. Used: inbound traffic enforcement at service edge, rate-limit specific applications. Sharp drop in application performance when limit hit. Shaping: enforces bandwidth by BUFFERING (queuing) excess traffic and sending later. Smooth traffic flow — no dropped packets (until buffer full). Used: outbound traffic before sending to slower WAN link. Shaping matches sending rate to the WAN link's committed rate — prevents drops on service provider side. Rule: police inbound, shape outbound.

Q157 What is DSCP to CoS mapping?

Ans Traffic crosses between Layer 3 (DSCP in IP header) and Layer 2 (CoS in 802.1Q tag) domains. Mapping needed: CoS 5 $\square$ DSCP EF (46) — VoIP. CoS 4 $\square$ DSCP AF41 (34) — video. CoS 3 $\square$ DSCP AF31 (26) — critical data. CoS 2 $\square$ DSCP AF21 (18) — standard data. CoS 1 $\square$ DSCP CS1 (8) — bulk. CoS 0 $\square$ DSCP 0 — best effort. Trust boundary: where the network trusts (or does not trust) QoS markings. Campus edge: trust CoS from IP phones, do not trust CoS from PCs (could mark traffic high priority to bypass QoS).

Q158 What is WRED (Weighted Random Early Detection)?

Ans WRED is a congestion avoidance mechanism that proactively drops some TCP packets before queues fill up. Why not wait for queue to fill? Full queue = tail drop = TCP global synchronisation (all TCP sessions back off simultaneously, then all restart = wave pattern). WRED: randomly drops a few packets from medium-priority traffic as queue grows. TCP detects loss, slows down gracefully. Different drop thresholds per DSCP value: EF never dropped, AF21 dropped earlier. Result: smoother traffic flow, avoids synchronisation. Configure: random-detect dscp-based under interface.

Q159 What is HSRP and how does preempt work?

Ans HSRP (Hot Standby Router Protocol) provides gateway redundancy — two routers share a virtual IP. Clients use virtual IP as default gateway. Active router handles traffic; Standby monitors. If Active fails: Standby takes over virtual IP — clients notice nothing (may have brief interruption). Priority: higher priority = Active role. Preempt: without preemption, when former Active router recovers it stays Standby — even if it has higher priority. With preempt: higher priority router reclaims Active role when it recovers. Configure: standby 1 priority 200 □ standby 1 preempt delay minimum 30 (wait 30 seconds before preempting — allows routing protocols to converge).

Q160 What is HSRP version 2 vs version 1?

Ans HSRP v1: 255 groups (0-255), uses 224.0.0.2 multicast, group timers in full seconds (minimum 1 second). v2: 4096 groups (0-4095), uses 224.0.0.102, millisecond timers (sub-second failover), includes VRRP timers compatibility, object tracking improvements. Configure: standby version 2. Both versions cannot interoperate on same group — all routers in group must use same version. Best practice: use HSRPv2 for all new deployments — millisecond timers enable sub-second failover when combined with BFD.

Q161 What is HSRP tracking?

Ans HSRP tracking links HSRP priority to the state of a tracked object — usually an interface or IP SLA. Example: router tracks its uplink to the internet. If uplink fails: reduce HSRP priority by 60 (from 200 to 140) □ other router (priority 150) becomes Active □ clients now send traffic through router with working internet path. Without tracking: Active router loses internet but keeps HSRP Active role — traffic black-holed. Configure: track 1 interface Gi0/1 line-protocol □ standby 1 track 1 decrement 60.

Q162 What is VRRP vs HSRP?

Ans HSRP: Cisco proprietary, Active/Standby, virtual IP separate from physical IPs, preempt disabled by default. VRRP (IEEE RFC 5798): open standard (works on Juniper, Arista, etc.), Master/Backup, virtual IP can be same as physical IP of master router (router that owns virtual IP has priority 255 automatically), preempt enabled by default. Functionally identical for most deployments. Choose VRRP: multi-vendor environment. Choose HSRP: Cisco-only + need Cisco-specific features (SSO integration). Configure VRRP: `vrrp 1 ip [virtual-IP] □ vrrp 1 priority 200`.

Q163 What is GLBP (Gateway Load Balancing Protocol)?

Ans GLBP provides true active-active gateway redundancy — unlike HSRP/VRRP where Standby is idle. GLBP assigns different virtual MAC addresses to different routers, all sharing the same virtual IP. AVG (Active Virtual Gateway): answers ARP requests for virtual IP. Distributes different virtual MACs to different clients (round-robin, weighted, host-dependent). AVF (Active Virtual Forwarder): each router is an AVF with its own virtual MAC. All routers actively forward traffic simultaneously. Configure: `glbp 1 ip [virtual-IP] □ glbp 1 weighting 200 □ glbp 1 load-balancing round-robin`.

Q164 What is NAT overloading (PAT) configuration?

Ans PAT (Port Address Translation) maps many private IPs to one public IP using unique source port numbers. Configure: `ip access-list standard NAT-ACL □ permit 192.168.0.0 0.0.255.255. ip nat inside source list NAT-ACL interface Gi0/1 overload`. Mark interfaces: `ip nat inside` (on LAN interface) + `ip nat outside` (on WAN interface). Verify: `show ip nat translations` (see active translations). `show ip nat statistics` (see hits, misses, total translations). Troubleshoot: `debug ip nat` (detailed translation events — use carefully in production).

Q165 What is static NAT?

Ans Static NAT maps one private IP permanently to one public IP — used for servers that need to be reachable from internet. Configuration: `ip nat inside source static 192.168.1.100 203.0.113.50`. Interface config same as PAT (`ip nat inside/outside`). With port translation (PAT static): `ip nat inside source static tcp 192.168.1.100 80 203.0.113.50 80` (only HTTP for this server). Verify: `show ip nat translations` (static entries always present). Common use case: internal web server, mail server, VPN concentrator that must be reachable via specific public IP.

Q166 What is IPv6 transition mechanisms?

Ans IPv6 transition mechanisms allow IPv4 and IPv6 coexistence: Dual stack: device runs both IPv4 and IPv6 simultaneously — most common enterprise approach. Tunnelling: encapsulate IPv6 in IPv4 (or vice versa). Types: 6in4 (manual tunnel), 6to4 (automatic, uses 2002::/16), ISATAP (intrasite), Teredo. 6PE: IPv6 over IPv4 MPLS backbone. Translation (NAT64): converts between IPv4 and IPv6 — for IPv6-only clients accessing IPv4 servers. DNS64: synthesises AAAA records from A records. For CCNP: focus on dual-stack and basic tunnelling concepts.

Q167 What is IPv6 NDP (Neighbour Discovery Protocol)?

Ans NDP replaces IPv4 ARP, ICMP redirect, and router discovery for IPv6. Uses ICMPv6 messages: Neighbour Solicitation (NS): 'Who has this IPv6 address?' (like ARP Request). Neighbour Advertisement (NA): 'I have this address, here is my MAC' (like ARP Reply). Router Solicitation (RS): client asks for router information. Router Advertisement (RA): router announces prefixes, default gateway, and flags. DAD (Duplicate Address Detection): before using an address, check no one else has it (send NS with target = own address). Verify: show ipv6 neighbors (equivalent to show arp).

Q168 What is IPv6 SLAAC and stateful DHCPv6?

Ans SLAAC (Stateless Address AutoConfiguration): device generates its own IPv6 address from router prefix + interface ID (from MAC address via EUI-64 or random). No DHCP server needed. Router sends RA with network prefix. Device combines prefix + interface ID. Stateful DHCPv6: like IPv4 DHCP — server assigns specific IPv6 addresses and DNS. Controlled by RA flags: M flag (Managed = use DHCPv6 for address). O flag (Other = use DHCPv6 for DNS/other info but SLAAC for address). If neither flag: use SLAAC only.

Q169 What is IP SLA?

Ans IP SLA (Service Level Agreement) generates synthetic test traffic to measure network performance. Types: ICMP echo (ping to measure latency/loss), UDP jitter (measure jitter and latency — critical for VoIP planning), TCP connect, HTTP (measure web response time). Configure: ip sla 1 ip udp-jitter 10.0.0.1 5004 codec g711alaw frequency 60 ip sla schedule 1 start-time now life forever. View results: show ip sla statistics 1 (see latency, jitter, loss). Use with object tracking for route failover and HSRP tracking.

Q170 What is multicast and how does PIM work?

Ans Multicast sends one stream to multiple interested receivers simultaneously — efficient for video streaming, IPTV, market data feeds. PIM (Protocol Independent Multicast) builds multicast distribution trees. Two modes: PIM-SM (Sparse Mode): source registers with Rendezvous Point (RP). Receivers join group via RP. Initial traffic through RP, then switches to Source Tree (SPT) for efficiency. PIM-DM (Dense Mode): flood then prune — simpler but wasteful. PIM-SSM (Source Specific Multicast): client specifies exact source IP — no RP needed, more secure. Configure: `ip multicast-routing` □ `ip pim sparse-mode` on interfaces □ `ip pim rp-address [RP-IP]`.

Q171 What is IGMP and its versions?

Ans IGMP (Internet Group Management Protocol) manages multicast group membership between clients and routers. IGMP v1: basic join, router querier. v2 (most common): adds leave messages (faster group leave), querier election, group-specific queries. v3: source-specific join — client can specify which source IP it wants multicast from (works with SSM). Router sends periodic IGMP queries to 224.0.0.1. Clients respond with membership reports for groups they want. IGMP snooping on switches: switch listens to IGMP and only sends multicast to ports with interested receivers — prevents flooding.

Q172 What is DHCP server, relay, and helper address?

Ans DHCP server: provides IP configuration to clients. Cisco router as server: `ip dhcp pool LAN` □ `network 192.168.1.0/24` □ `default-router 192.168.1.1` □ `dns-server 8.8.8.8`. Exclude static IPs: `ip dhcp excluded-address 192.168.1.1 192.168.1.20`. DHCP relay (helper address): DHCP uses broadcasts (not routed). Router must forward broadcasts to remote DHCP server. Configure on router interface facing clients: `ip helper-address [DHCP-server-IP]`. Router converts broadcast DHCP to unicast destined for DHCP server. Verify: `show ip dhcp binding`, `show ip dhcp statistics`.

Q173 What is NTP stratum and configuration?

Ans NTP (Network Time Protocol) synchronises device clocks. Stratum indicates accuracy level: Stratum 0 (atomic clock/GPS — reference). Stratum 1 (directly connected to Stratum 0). Stratum 2 (synchronised from Stratum 1) — this is what you configure routers to sync to. Each hop adds 1 stratum. Stratum 16 = unsynchronised. Configure: `ntp server 216.239.35.0 prefer` (Google NTP). Set timezone: `clock timezone IST 5 30` (India Standard Time, UTC+5:30). NTP authentication (optional): `ntp authentication-key 1 md5 MyKey` + `ntp trusted-key 1`. Verify: `show ntp status`, `show ntp associations`.

Q174 What is syslog and how do you configure it?

Ans Syslog sends device logs to a centralised server for storage, analysis, and correlation. Configure: logging host 192.168.1.100 (send to syslog server). logging trap informational (send informational and above). logging source-interface Loopback0 (use stable source IP). logging on (enable syslog, usually on by default). Local logging: logging buffered 65536 informational (keep logs in RAM buffer). View: show logging. Severity levels: 0 Emergency, 1 Alert, 2 Critical, 3 Error, 4 Warning, 5 Notice, 6 Informational, 7 Debugging. In production: log at least level 6 (informational) to remote server.

Q175 What is NetFlow and how is it configured?

Ans NetFlow exports IP traffic flow records (5-tuple: src/dst IP, src/dst port, protocol) to a collector for analysis. Uses: bandwidth billing, traffic analysis, capacity planning, security (detect DDoS, port scans, data exfiltration). Configure: interface Gi0/0 □ ip flow ingress (and/or egress). Specify collector: ip flow-export destination 192.168.1.100 9996 version 9. ip flow-export source Loopback0. Set active/inactive cache timers: ip flow-cache timeout active 1 (export active flows every 1 minute). Verify: show ip flow interface, show ip cache flow (sample records).

Q176 What is IPv6 OSPFv3?

Ans OSPFv3 is OSPF re-written for IPv6. Key changes: runs over IPv6 link-local addresses (not physical IPv4 IPs), configured per-interface (not using network statements), new LSA types (8=Intra-Area Prefix, 9=Inter-Area Prefix), authentication via IPv6 IPsec (not OSPF-native auth). Two configuration models: Classic (ipv6 ospf 1 area 0 on interface, separate from OSPFv2). Modern address-family model (OSPFv3 supports both IPv4 and IPv6 via address families — configure under router ospfv3 process □ address-family ipv4/ipv6). Show: show ospfv3 neighbor, show ospfv3 database.

Q177 What is first-hop security?

Ans First-hop security protects clients from Layer 2 attacks. Suite: DHCP Snooping (prevents rogue DHCP servers — covered in L2 section). Dynamic ARP Inspection (DAI): validates ARP packets against DHCP snooping binding table — prevents ARP poisoning. IPv6 RA Guard: prevents rogue IPv6 router advertisements (equivalent of DHCP snooping for IPv6). IPv6 DHCP Guard: prevents rogue DHCPv6 servers. IPv6 Source Guard: filters IPv6 traffic based on binding table. SEND (Secure NDP): cryptographic protection for NDP. In CCNP: focus on DHCP snooping, DAI, and RA Guard conceptually.

Q178 What is SNMP v3 configuration?

Ans SNMPv3 provides authentication and encryption — critical for production networks. Three security levels: noAuthNoPriv (no auth, no privacy). authNoPriv (MD5/SHA auth, no encryption). authPriv (auth + AES encryption — use this). Configure: `snmp-server group ADMIN v3 priv (create group)`. `snmp-server user akhilesh ADMIN v3 auth sha MyAuthPass priv aes 128 MyPrivPass`. `snmp-server host 192.168.1.100 version 3 priv akhilesh (send traps)`. Verify: `show snmp user`, `show snmp group`. Enable specific traps: `snmp-server enable traps bgp`, `snmp-server enable traps ospf`.

Q179 What is EEM (Embedded Event Manager)?

Ans EEM allows network devices to automatically respond to events — self-healing automation. Event detectors: syslog messages, interface up/down, CLI commands typed, timer, SNMP threshold, track object state changes. Actions: run CLI commands, send email, generate SNMP trap, run Tcl/Python script, reload interface. Example applet: when 'BGP_CONN_RESET' syslog occurs □ execute 'clear ip bgp soft' + send syslog 'BGP reset detected, auto-recovering'. Configure: `event manager applet BGP-RECOVER` □ `event syslog pattern 'BGP_CONN_RESET'` □ `action 1.0 cli command 'clear ip bgp 10.0.0.1 soft'`. Very powerful for automated operations.

Q180 What is LIMP and IP SLA for tracking?

Ans Object tracking combined with IP SLA enables automated route and HSRP changes. IP SLA generates probes. Tracking object monitors IP SLA results. HSRP or static routes react to tracking object state. Example: probe ISP-A internet path with IP SLA □ track object monitors if probe succeeds □ if probe fails: tracking object goes down □ static route via ISP-A is removed □ traffic switches to ISP-B □ HSRP decrements priority □ secondary router becomes HSRP Active. Full automation without human intervention. Configure: `ip sla 1` □ `track 1 ip sla 1 reachability` □ `ip route 0.0.0.0 0.0.0.0 [ISP-A] track 1`.

SECTION 07

Security & VPN

Q181 –
Q210

ACLs · Zone-Based Firewall · IPsec VPN · DMVPN · GRE · PKI · AAA · RADIUS

Q181 What is a Zone-Based Firewall (ZBFW) on IOS?

Ans ZBFW is a stateful firewall implemented directly on a Cisco router (IOS 12.4(6)T+). Zones: logical groups of interfaces with same security level (inside, outside, DMZ). Policies between zones define allowed traffic. By default: traffic between different zones is BLOCKED. Within same zone: allowed. Traffic to router itself (self zone): controlled by policy from zone to self. Benefits over ACLs: stateful (tracks TCP sessions, allows return traffic automatically), application inspection (inspect FTP, H.323, SIP), easy to understand zone model.

TIP ZBFW is directly tested in CCNP – understand zone configuration and policy maps!

Q182 How do you configure ZBFW on Cisco IOS?

Ans Step-by-step: 1) Create zones: zone security INSIDE, zone security OUTSIDE, zone security DMZ. 2) Assign interfaces: zone-member security INSIDE on LAN interface. 3) Create class-maps (match traffic): class-map type inspect match-any WEB □ match protocol http. 4) Create policy-map (actions): policy-map type inspect INSIDE-TO-OUTSIDE □ class WEB □ inspect. 5) Create zone-pair (apply policy): zone-pair security INSIDE-TO-OUTSIDE source INSIDE destination OUTSIDE □ service-policy type inspect INSIDE-TO-OUTSIDE. Verify: show policy-map type inspect zone-pair, show zone security.

Q183 What is an extended ACL and where do you place it?

Ans Extended ACLs filter traffic based on: source IP, destination IP, protocol (TCP/UDP/ICMP), source port, and destination port. Syntax: ip access-list extended MY-ACL □ permit tcp 192.168.1.0 0.0.0.255 any eq 80 (permit HTTP from 192.168.1.x to anywhere). Placement rule: place extended ACLs as CLOSE TO THE SOURCE as possible – stop traffic early before it wastes bandwidth traversing the network. Inbound on source interface stops traffic immediately. Standard ACLs (source IP only): place close to destination – because they cannot match destination, placing near source may block too broadly.

Q184 What is ZBFW application inspection?

Ans Application inspection understands application protocols at Layer 7 and dynamically opens firewall pinholes for them. Example: FTP uses port 21 for control but negotiates a separate data connection on dynamic port — firewall must open this dynamic port. Without inspection: data connection blocked (only port 21 allowed). With inspection: firewall tracks FTP control session, detects PORT/PASV command, automatically opens required data port. ZBFW inspect parameter: class WEB □ inspect (enables stateful inspection + application inspection). Protocols: H.323, SIP, FTP, TFTP, SQL*Net, etc.

Q185 What is Cisco IOS IPsec VPN?

Ans IPsec provides secure site-to-site VPN. Two phases: IKE Phase 1 (ISAKMP): authenticates peers and establishes encrypted channel for Phase 2 negotiations. Uses: DH key exchange, pre-shared key or certificate authentication. IKE Phase 2 (IPsec SA): negotiates encryption for actual traffic. Parameters: encryption algorithm (AES-256), hash (SHA-256), lifetime, traffic to protect (crypto ACL). Configuration: crypto isakmp policy 10 □ crypto isakmp key PSK-KEY address [peer-IP] □ crypto ipsec transform-set MY-TSET esp-aes 256 esp-sha-hmac □ crypto map MY-MAP 10 ipsec-isakmp.

Q186 What is GRE tunnel and when do you use it with IPsec?

Ans GRE (Generic Routing Encapsulation) creates a virtual tunnel between routers — supports routing protocols (OSPF/EIGRP over tunnel) and multicast. GRE alone: no encryption. IPsec alone: encrypts traffic but no dynamic routing (only specific traffic defined in crypto ACL). GRE over IPsec: run OSPF/EIGRP over GRE tunnel + IPsec encrypts the GRE tunnel. Best of both worlds: dynamic routing + encryption. Configuration: tunnel interface with tunnel mode gre ip + tunnel protection ipsec profile MY-PROFILE. The GRE tunnel interface is configured with IP address and OSPF runs over it.

Q187 What is DMVPN (Dynamic Multipoint VPN)?

Ans DMVPN creates a scalable hub-and-spoke VPN where spokes can build direct tunnels to each other on demand. Traditional hub-and-spoke: spoke-to-spoke traffic always goes through hub (double WAN hop). DMVPN: spoke-to-spoke traffic triggers NHRP (Next Hop Resolution Protocol) to resolve other spoke's real IP □ direct IPsec tunnel built on demand. Components: GRE multipoint interface (mGRE) on hub and spokes, NHRP for IP resolution, IPsec for encryption. Phases: Phase 1 (spoke-to-hub only), Phase 2 (spoke-to-spoke triggered by traffic), Phase 3 (optimised Phase 2). Used by enterprises with many branches.

Q188 What is DMVPN Phase 3?

Ans DMVPN Phase 3 improves on Phase 2 by using NHRP redirect messages. Phase 2: spoke must know hub's routing before building direct tunnel — complex routing. Phase 3: hub sends NHRP Redirect to spoke — 'reach that destination directly, here is the other spoke's NBMA address'. Spoke builds direct tunnel. Advantages: cleaner routing (spokes do not need specific routes for each other), summarisation possible at hub (hubs can advertise summaries, spokes still build direct tunnels for specifics), faster tunnel establishment. Configure: tunnel source-nhrp redirect (hub) + tunnel destination 0.0.0.0 (spokes).

Q189 What is FlexVPN?

Ans FlexVPN is Cisco's modern VPN framework using IKEv2 (Internet Key Exchange version 2). IKEv2 advantages over IKEv1: fewer message exchanges (faster setup), built-in NAT traversal, dead peer detection, supports MOBIKE (client IP change without renegotiation). FlexVPN supports: site-to-site (like DMVPN), remote access (IKEv2 with Cisco AnyConnect), spoke-to-spoke. Configuration uses Virtual Tunnel Interface (VTI) — cleaner than crypto maps. FlexVPN replacing DMVPN in new deployments. Also supports: EAP authentication, certificate authentication, radius/LDAP integration.

Q190 What is Cisco TrustSec?

Ans TrustSec is Cisco's scalable group-based access control framework (same as SD-Access SGTs). Assigns Security Group Tags (SGTs) to users and devices based on identity. SGACLs (Security Group ACLs) define allowed communication between SGT pairs — independent of IP or VLAN. SGT propagation: inline tagging (Cisco switches tag frames with SGT at ingress) or SXP (SGT Exchange Protocol — for older switches that cannot tag). ISE assigns SGTs via RADIUS. TrustSec enables: microsegmentation, consistent policy across entire network, policy remains with user regardless of location.

Q191 What is AAA (Authentication, Authorisation, Accounting)?

Ans AAA provides centralised security management. Authentication: verify identity — who are you? (username/password, certificate, MFA). Authorisation: what are you allowed to do? (which CLI commands, which resources). Accounting: what did you do? (audit trail — log every command, session, bandwidth used). Implement with RADIUS or TACACS+:
aaa new-model
aaa authentication login default group radius local (try RADIUS first, then local).
aaa authorisation exec default group tacacs+ local (command authorisation via TACACS+).
aaa accounting commands 15 default start-stop group tacacs+ (log all privileged commands).

Q192 What is the difference between RADIUS and TACACS+?

Ans RADIUS (RFC 2865/2866): open standard, UDP 1812 (auth) / 1813 (acct), combines authentication + authorisation, encrypts only the password. Preferred for: network access (802.1X, VPN, Wi-Fi users). Attribute-based: RADIUS attributes control access (VLAN assignment, ACL download). TACACS+ (Cisco proprietary): TCP 49, separates authentication/authorisation/accounting (more granular), encrypts entire packet. Preferred for: network device administration (router/switch login, command authorisation). TACACS+ can control specific CLI commands — RADIUS cannot. Enterprise: use RADIUS for users, TACACS+ for network device management.

Q193 What is ZBFW with threat-defense (NGFW)?

Ans Cisco IOS ZBFW is a basic firewall. Cisco's Next-Generation Firewall (NGFW) — Firepower Threat Defense (FTD) — provides: Application Visibility and Control (AVC): identify and control 3000+ applications by name. URL Filtering: block by URL category (gambling, social media). IPS (Intrusion Prevention System): detect and block network exploits. Malware protection (AMP): detect malware in files. SSL inspection: decrypt and inspect HTTPS. All managed by FMC (Firepower Management Center). CCNP ENCOR: understand FTD/NGFW concepts, not deep configuration.

Q194 What is PKI (Public Key Infrastructure) in VPN context?

Ans PKI provides certificate-based authentication for VPNs — more scalable than pre-shared keys (PSK). With PSK: need unique key per site pair — 100 sites = 4950 keys! With PKI: each device gets a certificate from a trusted CA (Certificate Authority). Peer authentication: 'I trust your certificate because our shared CA signed it.' CA types: Internal CA (IOS Certificate Server or Microsoft CA), External CA (public — DigiCert, Comodo). Configuration: `crypto pki trustpoint MY-CA` □ `enrollment url [CA-URL]` □ `subject-name CN=[hostname]`. Request certificate: `crypto pki enroll MY-CA`.

Q195 What is Cisco AnyConnect VPN?

Ans Cisco AnyConnect is an SSL/TLS and IKEv2 remote access VPN client. Features: SSL/TLS VPN (works through any firewall — uses HTTPS port 443), IKEv2/IPsec (better performance, more features), DART (Diagnostic and Reporting Tool — collects logs for troubleshooting), Network Access Manager (802.1X), Host Scan (checks client security posture before granting access), ISE Posture (integration with ISE for compliance checking). Server: Cisco ASA, Cisco FTD, or IOS router. CCNP: understand AnyConnect vs clientless SSL VPN differences.

Q196 What is a clientless SSL VPN?

Ans Clientless SSL VPN (also called WebVPN) provides access to internal resources through a web browser — no client software installation. Users navigate to the VPN gateway URL (<https://vpn.company.com>), authenticate, and access: internal web applications, file shares (SMB via web), email, published applications. Advantages: no client installation (good for BYOD), works anywhere with a browser. Disadvantages: limited application support (only web-based), poor experience for non-browser apps, security limitations. AnyConnect (full client) provides better experience — clientless is for occasional or BYOD access.

Q197 What is Crypto ACL in IPsec VPN?

Ans Crypto ACL (interesting traffic ACL) defines which traffic to encrypt through an IPsec tunnel. Configuration: `ip access-list extended CRYPTO-ACL` `permit ip 192.168.1.0 0.0.0.255 10.0.0.0 0.0.255.255`. Applied in crypto map: `crypto map MY-MAP 10` `match address CRYPTO-ACL`. Traffic matching ACL: encrypted and sent through tunnel. Non-matching traffic: sent normally (internet traffic goes direct if using split tunnelling). Important: crypto ACLs on both tunnel endpoints must be mirrors of each other — what one side permits to encrypt, other side must permit to decrypt.

Q198 What is IKEv2 vs IKEv1?

Ans IKEv1: RFC 2409, two modes (Main mode — 6 messages, Aggressive mode — 3 messages but less secure), separate XAUTH for user authentication, limited NAT traversal. IKEv2: RFC 4306/7296, 4 messages (much faster — one round trip for Phase 1 + Phase 2), built-in NAT traversal (NAT-T), EAP authentication support (username/password + MFA), MOBIKE (handle IP changes without tunnel renegotiation), more cryptographic agility. FlexVPN uses IKEv2. All new VPN deployments should use IKEv2. Verify: `show crypto ikev2 session`.

Q199 What is GETVPN (Group Encrypted Transport VPN)?

Ans GETVPN encrypts traffic natively within a network — without point-to-point tunnels. Traditional IPsec: each site-pair needs a tunnel. GETVPN: encryption policy distributed to all routers from a Key Server (using GDOI protocol). All routers share the same encryption keys. Traffic between GETVPN members: encrypted in-place, routed normally. Benefits: any-to-any encrypted connectivity, maintains native IP routing (no overlay), MPLS QoS markings preserved (unlike GRE-based VPN). Used by enterprises with full-mesh MPLS — want encryption without re-designing routing. Limited to trusted provider networks (no internet transport).

Q200 What is CoPP (Control Plane Policing)?

Ans CoPP protects the router's control plane (CPU) from being overwhelmed by excessive traffic. Without CoPP: an attacker can flood the router CPU with ICMP pings or BGP packets — routing protocols fail, device becomes unmanageable. CoPP: rate-limits traffic destined for the router's own CPU. Policy: allow unlimited OSPF hellos from known neighbours, allow BGP at 100 Mbps, rate-limit ICMP to 1000 pps, drop unknown protocol traffic. Configure with MQC: policy-map COPP-POLICY □ class ICMP □ police 1000000 □ apply with control-plane □ service-policy input COPP-POLICY. Essential for internet-connected routers.

Q201 What is uRPF (Unicast Reverse Path Forwarding)?

Ans uRPF is an anti-spoofing feature. It checks: does the source IP of an incoming packet have a valid route back out the same interface it arrived on? If not: likely spoofed — drop the packet. Two modes: Strict mode (the back-route must be via the same interface) — use on single-homed interfaces. Loose mode (back-route can be via any interface) — use on asymmetric routing paths. Configure: ip verify unicast source reachable-via rx (strict) or any (loose). Important for internet edge routers — prevents DDoS amplification using spoofed source IPs.

Q202 What is prefix hijacking and RPKI?

Ans BGP prefix hijacking: a malicious or misconfigured router advertises another organisation's IP prefixes — attracting traffic destined for the victim. Famous example: Pakistan Telecom accidentally announced YouTube's prefixes — YouTube unreachable globally for 2 hours (2008). RPKI (Resource Public Key Infrastructure): cryptographically validates that a BGP announcement is authorised by the prefix owner. Route Origin Authorisation (ROA): prefix owner signs a record at their RIR saying 'AS 64500 is authorised to announce 192.0.2.0/24'. BGP routers with RPKI validation: mark invalid ROAs as invalid, prefer valid routes. Increasingly deployed by major ISPs.

Q203 What is HSRP authentication?

Ans HSRP authentication prevents rogue routers from joining the HSRP group. Without authentication: any router sending higher-priority HSRP hellos becomes Active — traffic black-holed to untrusted router. Two methods: Plain text (weak): standby 1 authentication cisco123. MD5 (recommended): standby 1 authentication md5 key-string MyHSRKey. Key chains: standby 1 authentication md5 key-chain HSRP-KEY. Both Active and Standby must have same authentication. Mismatched authentication: routers cannot hear each other's BPDUs — two Active routers (both think they are Active = split brain).

Q204 What is EIGRP MD5 vs HMAC-SHA-256 authentication?

Ans MD5 authentication (classic EIGRP): configure key chain, apply to interface. Works but MD5 is considered cryptographically weak by modern standards. HMAC-SHA-256 (EIGRP Named Mode): stronger modern authentication. Configure under address-family: `router eigrp MY-INSTANCE` `address-family ipv4 unicast` `as 1` `af-interface Gi0/0` `authentication mode hmac-sha-256 MyKey`. Both peers must use same authentication mode — cannot mix MD5 and HMAC-SHA-256. For new deployments: use EIGRP Named Mode with HMAC-SHA-256. For existing MD5 deployments: plan migration during maintenance window.

Q205 What is BGP maximum-prefix limit?

Ans BGP maximum-prefix is a safety mechanism preventing routing table overflow from a misbehaving peer. Configure: `neighbor 203.0.113.1 maximum-prefix 750000 80`. 750000 = tear down session if peer sends more than 750,000 prefixes. 80 = generate warning at 80% (600,000 prefixes). Restart timer: `neighbor 203.0.113.1 maximum-prefix 750000 80 restart 5` (automatically try to re-establish session after 5 minutes). Essential for internet routers receiving full BGP table (~900,000 routes as of 2024). Without this: memory exhaustion can crash the router.

Q206 What is keychain rotation in IOS authentication?

Ans Key chains support multiple keys with send and accept lifetimes — enabling graceful key rotation without service disruption. Example: `key chain OSPF-KEYS` `key 1` `key-string OldPassword` `accept-lifetime infinite` `send-lifetime 2024-01-01 00:00:00 2024-07-01 00:00:00`. `key 2` `key-string NewPassword` `accept-lifetime 2024-06-01 00:00:00 infinite` `send-lifetime 2024-07-01 00:00:00 infinite`. Overlap period: both keys valid for accept (receive) but only old key sent until cutover. After cutover: only new key sent. Both routers update at same time without disruption.

Q207 What is Cisco Flexible Packet Matching (FPM)?

Ans FPM allows IOS to match and take actions on arbitrary bytes within packet payloads — beyond what standard ACLs can match. Use cases: blocking specific patterns in application data, matching protocols not recognised by standard protocol keyword, custom deep packet inspection. Configuration uses class-maps with match field patterns. Limited use in modern networks (typically replaced by NGFW solutions) but still relevant for specific ISP and carrier use cases. CCNP: understand FPM concept and what problems it solves rather than detailed configuration.

Q208 What is a site-to-site VPN with certificate authentication?

Ans Certificate-based IPsec VPN: more secure and scalable than pre-shared keys. Each router has: its own certificate (signed by CA), the CA's certificate installed as trusted root. IKE Phase 1 authentication: each router presents certificate □ peer verifies against trusted CA □ identity confirmed. No need for shared secrets between peers. Configuration: `crypto pki trustpoint MY-CA` □ `install CA cert` □ `enroll router` □ `crypto isakmp profile MY-PROFILE` □ `ca trust-point MY-CA` □ `match identity address [peer-IP]` □ `initiate-mode aggressive`. Verify: `show crypto isakmp profile`.

Q209 What is TACACS+ command authorisation?

Ans TACACS+ can authorise specific CLI commands — define exactly which commands each user can type. Configuration: `aaa authorisation commands 15 default group tacacs+ none` (authorise privilege-15 commands via TACACS+). On TACACS+ server (ISE, CiscoACS): create authorisation profile. Permit: user admin can run all commands. Restrict: user helpdesk can only run show commands. Deny: user readonly cannot run clear or debug. Important: configure local fallback (none at end of method list) — if TACACS+ server unreachable, no authorisation check. Without fallback: complete lockout if TACACS+ server fails.

Q210 What is Flexible NetFlow (FNF)?

Ans Flexible NetFlow (FNF) improves on traditional NetFlow by allowing you to define exactly what fields to capture. Traditional NetFlow: fixed 5-tuple (src/dst IP, port, protocol). FNF: choose any combination of fields: IPv4 fields, IPv6 fields, Layer 2 fields, MPLS labels, application IDs, timestamps. Create flow record (what fields to capture): `flow record MY-RECORD` □ `match ipv4 destination address` □ `match transport destination-port` □ `collect counter bytes`. Create flow exporter. Create flow monitor (ties record to exporter). Apply to interface: `ip flow monitor MY-MONITOR input`. Verify: `show flow monitor MY-MONITOR cache`.

SECTION 08

Infrastructure Services & Management

Q211 –
Q240

SNMP · EEM · Automation · Python · Ansible · RESTCONF · NETCONF · DNA Center

Q211 What is model-driven programmability in IOS XE?

Ans Model-driven programmability uses structured data models (YANG) to configure and monitor IOS XE devices programmatically. Instead of parsing text output from show commands: tools request structured data. Protocols: NETCONF (SSH, XML, port 830), RESTCONF (HTTPS, JSON/XML, port 443), gRPC (streaming telemetry). Benefits: automation without screen-scraping, vendor-neutral models (OpenConfig), machine-readable configuration, consistent APIs across all IOS XE devices. Enable: netconf-yang and restconf (under configure terminal). CCNP requires understanding of this architecture.

TIP*Automation and programmability is ~15% of CCNP ENCOR – do not skip it!***Q212** What is YANG and why is it important?

Ans YANG (Yet Another Next Generation) is a data modelling language that defines the structure of network device configuration and state data. Like a schema for network data. Example: a YANG model defines that a router has interfaces, each interface has a name (string), IP address, enabled state (boolean). NETCONF and RESTCONF use YANG models. Benefits: machine-readable interface, cross-vendor consistency (OpenConfig YANG models work on Cisco, Juniper, Arista). Cisco IOS XE YANG models published on GitHub. Tools like pyang validate YANG models.

Q213 What is NETCONF and how do you use it?

Ans NETCONF (RFC 6241) is a standardised protocol for programmatically managing network devices. Uses XML over SSH (port 830). Four operations: get (retrieve config/state), edit-config (change configuration), copy-config (copy between datastores), lock/unlock (prevent concurrent changes). IOS XE datastores: running (active config), startup (saved config), candidate (working copy – not all platforms). Enable: netconf-yang. Python library: ncclient. Example: device.edit_config('..set hostname.xml..', target='running'). Output: XML-formatted configuration. Used by NSO, Ansible, custom scripts.

Q214 What is RESTCONF?

Ans RESTCONF (RFC 8040) provides REST API access to the same YANG data models as NETCONF but using HTTP/HTTPS (port 443) and JSON/XML. Standard HTTP verbs: GET (retrieve), POST (create), PUT (replace), PATCH (update), DELETE (remove). Base URL: `https://[router-IP]/restconf/data/`. Example: GET `https://10.0.0.1/restconf/data/ietf-interfaces:interfaces` — returns all interface configuration in JSON. Much easier to use than NETCONF (curl, Postman, Python requests). Enable: `restconf`. Authentication: basic auth or token. Verify: GET `https://10.0.0.1/restconf/yang-library-version`.

Q215 What is streaming telemetry on IOS XE?

Ans Streaming telemetry (model-driven telemetry) continuously pushes operational data from the device to a collector — not polling like SNMP. Protocols: gRPC (port 57500 — Cisco native), gNMI (gRPC Network Management Interface — open standard), NETCONF subscriptions. Data pushed: interface counters (every 30 seconds), BGP session state, CPU/memory, routing changes. Benefits: real-time visibility, detect brief events SNMP polling misses, push from device (not pull = lower device CPU), high-frequency data (every second if needed). Tools: Telegraf (collect) + InfluxDB (store) + Grafana (visualise) — open-source stack.

Q216 What is Python network automation?

Ans Python is the standard language for network automation. Key libraries: netmiko (SSH connections to network devices — execute show commands, push config), ncclient (NETCONF), requests (REST API calls — RESTCONF, vManage, DNA Center), textFSM (parse show command text into structured data), napalm (multi-vendor abstraction layer). Simple workflow: `import netmiko`, define device dict (host, username, password, device_type), `connection = ConnectHandler(**device)`, `output = connection.send_command('show ip route')`, `print(output)`. Process output with regex or textFSM to extract specific data.

Q217 What is Ansible for network automation?

Ans Ansible automates network device configuration using agentless connections (SSH or API). Playbooks: YAML files describing desired state. Inventory: list of devices. Modules: `cisco.ios.ios_config` (push IOS config), `cisco.ios.ios_command` (run show commands), `cisco.ios.ios_bgp_global` (BGP config). Example task: `cisco.ios.ios_config` ▯ lines: `['router ospf 1', 'network 192.168.0.0 0.0.0.255 area 0']`. Idempotent: running playbook twice does not cause duplicate config (checks current state first). Version control: store playbooks in Git — change history, peer review, rollback.

Q218 What is Terraform for network infrastructure?

Ans Terraform (HashiCorp) manages infrastructure as code — declarative (define desired state, Terraform figures out how to achieve it). Network providers: cisco-ios, cisco-iosxe, Cisco DNA Center, vManage SD-WAN, AWS/Azure networking. Define in HCL (HashiCorp Configuration Language): resource iosxe_bgp_neighbor { remote_as = 65002, ip = 10.0.0.2 }. Terraform apply: reads current state, calculates diff, applies changes. Terraform state: tracks what was deployed. Terraform destroy: removes all managed resources. Used for: cloud networking, SD-WAN deployments, consistent multi-site configurations.

Q219 What is Cisco NSO (Network Services Orchestrator)?

Ans Cisco NSO (formerly Tail-f NCS) is an enterprise network automation platform. Features: multi-vendor support (Cisco IOS, NX-OS, IOS XR, Juniper, Ericsson, Nokia), service modelling (define 'L3VPN service' template — NSO generates all device configs), transaction manager (commit all changes atomically — either all succeed or none), rollback (full network rollback to previous state), compliance reporting (compare running vs intended config). NSO uses NETCONF/YANG for modern devices, CLI adapters for legacy. Most large ISPs use NSO for service provisioning.

Q220 What is Cisco DNA Center REST API?

Ans DNA Center exposes all functionality via REST API. Authentication: POST https://[DNAC-IP]/dna/system/api/v1/auth/token (get JWT token). Common API calls: GET /dna/intent/api/v1/network-device (list all devices), GET /dna/intent/api/v1/issues (get current issues), POST /dna/intent/api/v1/template-programmer/template (create config template), GET /dna/intent/api/v1/client-detail?macAddress=xx:xx:xx:xx:xx:xx (client 360 data). Authentication header: x-auth-token: [JWT token]. Useful for: integration with ServiceNow (auto-ticket creation), custom dashboards, automated provisioning.

Q221 What is EEM (Embedded Event Manager) scripting?

Ans EEM scripts can range from simple applets (inline Tcl in config) to complex Tcl/Python scripts. Simple applet example: event manager applet LINK-DOWN □ event syslog pattern 'LINK-5-CHANGED.*down' □ action 1 syslog msg 'Link down detected — running diagnostics' □ action 2 cli command 'show interfaces' □ action 3 mail server 10.0.0.100 to noc@company.com subject 'Link Down Alert' body 'Interface down' □ action 4 counter name LINK-FLAP increment 1. Python EEM: event manager applet PY-SCRIPT □ action 1 cli command 'guestshell run python3 /flash/myscript.py'.

Q222 What is Guestshell on IOS XE?

Ans Guestshell is a Linux container (CentOS) running on IOS XE platforms — provides a Python 3 environment for on-device automation. Benefits: run Python scripts directly on the router, access Cisco device libraries (cli module to run IOS commands from Python), no external automation server needed. Enable: guestshell enable. Access: guestshell run bash or guestshell run python3. Import Cisco CLI module: import cli; output = cli.execute('show ip route'). Use with EEM: trigger Python script from EEM when events occur. Persistent scripts: copy to flash, run from EEM or cron-like scheduler.

Q223 What is SD-WAN automation with vManage API?

Ans vManage exposes comprehensive REST API. Authentication: POST /j_security_check (cookie-based auth). Common calls: GET /dataservice/device (all SD-WAN devices), GET /dataservice/device/bfd/summary (BFD status), POST /dataservice/template/device/config/attachfeature (attach config template to device), GET /dataservice/statistics/interface (interface statistics). Python example: session = requests.Session() □ POST login □ GET device list □ parse JSON response. Used for: automated ZTP, performance monitoring, integration with ITSM tools, custom dashboards showing SD-WAN health.

Q224 What is textFSM and why do network engineers use it?

Ans textFSM is a Python library for parsing semi-structured CLI text output into structured data. Problem: show ip bgp summary returns text — hard to process in code. Solution: textFSM template defines regex patterns for each field (Neighbor, AS, State, Prefixes). Library parses output against template □ returns list of dicts: [{'Neighbor': '10.0.0.1', 'AS': '65002', 'State': 'Established', 'Prefixes': '123456'}]. Ntc-templates: library of pre-built textFSM templates for common Cisco/Juniper/Arista commands — parse show outputs without writing regex. import netmiko + ntc_templates = powerful automation without NETCONF.

Q225 What is NAPALM (Network Automation and Programmability)?

Ans NAPALM provides a unified API for managing multiple vendor devices (Cisco IOS, IOS XE, NX-OS, Juniper JunOS, Arista EOS). Same Python code works across vendors. Functions: `get_interfaces()`, `get_bgp_neighbors()`, `load_merge_candidate()` (merge config changes), `load_replace_candidate()` (replace full config), `commit_config()`, `discard_config()`, `rollback()`. Especially powerful: open-source, uniform API, handles vendor differences internally. Used in: multi-vendor deployments, compliance checking scripts, configuration management tools.

Q226 What is Cisco CML (Cisco Modelling Labs)?

Ans Cisco Modelling Labs (formerly VIRL) is Cisco's official network simulation platform — runs real IOS XE, IOS XR, NX-OS images in virtual machines. Used for: CCNP/CCIE lab practice without physical hardware, automated testing of network changes before production deployment, creating network topologies to test automation scripts. Commercially licensed (per node license). CML 2.x: web-based interface, REST API, integration with Ansible and CI/CD pipelines. Free option: Cisco DevNet community edition (limited node count). Also: EVE-NG (free community edition) supports same Cisco images.

Q227 What is CI/CD for network automation?

Ans CI/CD (Continuous Integration / Continuous Delivery) applies software development practices to network changes. Workflow: engineer pushes config change to Git repository □ CI pipeline triggers automatically □ Ansible syntax check □ lint check □ test against lab (CML) □ if tests pass □ automated production deployment via approval gate. Benefits: every change is tested, reviewed (peer review via pull request), traceable (Git history), and reversible. Prevents 'works in my lab but breaks production' scenarios. Tools: GitLab CI, GitHub Actions, Jenkins, plus Ansible, Terraform, Batfish (network configuration validation).

Q228 What is Batfish for network validation?

Ans Batfish is an open-source network analysis tool that validates network configurations before deployment — without running them on real hardware. Takes configuration files □ builds a complete network model □ answers questions: 'Will OSPF adjacency form between these two routers?', 'What will happen to traffic if this link fails?', 'Are ACLs configured correctly?', 'Verify change does not break reachability'. Catches bugs that human review misses. Integrate into CI/CD pipeline: every configuration commit automatically validated against network policies. Very powerful for large-scale change management.

Q229 What is pyATS (Python Automated Test Systems)?

Ans pyATS is Cisco's Python testing framework for network validation. Combines: pyATS (test framework) + Genie (libraries for network device interaction — structured output parsers, testcases). Use cases: baseline testing (capture network state before and after change), regression testing, automated troubleshooting. Example: connect to device, run 'learn ospf' (Genie captures OSPF neighbours), make a change, run 'learn ospf' again, compare (diff) □ automatically detects any OSPF changes. Very used in network DevOps pipelines for change validation.

Q230 What is Cisco DevNet and certification?

Ans Cisco DevNet (developer.cisco.com) is Cisco's developer community for network automation and programmability. Resources: Sandbox environments (free virtual lab access — DNA Center, vManage, IOS XE, ACI, etc.), Learning Labs (step-by-step programming guides), API documentation, open-source code samples. Certifications: DevNet Associate (DEVASC 200-901), DevNet Professional (DEVCOR 350-901), DevNet Specialist (ENAUTO 300-435 — Enterprise Automation — directly relevant to CCNP). CCNP Enterprise candidates interested in automation should consider ENAUTO specialisation.

Q231 What is SNMP traps vs SNMP polling?

Ans Polling: NMS sends GET requests to device every N minutes — requests specific OIDs (CPU, interface counters). Simple but misses brief events between polls. Trap: device proactively sends notification to NMS when specific event occurs (interface down, BGP session reset, CPU threshold exceeded). Immediate notification — does not wait for next poll cycle. Best practice: use BOTH. Polling for trend data (bandwidth utilisation graphs). Traps for immediate alerts (interface down alarm). SNMPv3 trap (Inform): trap with acknowledgement — device retries if NMS does not acknowledge (standard trap has no acknowledgement).

Q232 What is Cisco DNA Spaces?

Ans Cisco DNA Spaces is a cloud-based location and analytics platform. Collects Wi-Fi signal data from Cisco WLCs and Meraki — tracks connected devices and visitors in physical spaces. Use cases: occupancy analytics (how many people in meeting room?), visitor journey (which areas they visited), dwell time, asset tracking (locate specific device or equipment), proximity notifications (alert when device enters restricted area). Data sent to DNA Spaces cloud platform. Integrates with DNA Center. Retail, healthcare, campus, and enterprise building management use cases.

Q233 What is Intent-Based Networking (IBN)?

Ans Intent-Based Networking (IBN) translates high-level business intent into network configuration — the network itself figures out the technical details. Traditional: 'configure OSPF area 0 on interface Gi0/0 with cost 10'. IBN: 'connect engineering floor to server farm with 10Gbps guarantee'. DNA Center translates intent to specific device configuration. Components: Translation (intent to policy), Activation (push policy to devices), Assurance (verify intent is implemented correctly, continuously). IBN closes the loop: if network drifts from intent — automatically corrects or alerts. Cisco DNA Center is Cisco's IBN platform.

Q234 What is Zero-Trust Architecture?

Ans Zero Trust: 'Never trust, always verify' — no implicit trust based on network location (inside the firewall does not mean trusted). Principles: verify explicitly (authenticate and authorise every access request — MFA, device compliance), least-privilege access (minimum necessary access per request), assume breach (segment network to limit blast radius). Implementation: Identity (Azure AD / ISE — who is this?), Device (MDM compliance — is this device healthy?), Network (SGACL — what can this device access?), Application (ZTNA — per-application access rather than broad VPN). Cisco: SD-Access + ISE + Duo (MFA) + Umbrella (DNS security).

Q235 What is Cisco SecureX?

Ans Cisco SecureX is Cisco's cloud-based security integration platform — connects Cisco security products (Firepower, Umbrella, AMP, ISE, SecureEndpoint) into a unified dashboard. Features: threat intelligence aggregation (see all security events in one place), automated investigation (AI-correlates events across products), response orchestration (automated remediation workflows), casebooks (document and track investigations). Reduces analyst work: one alert might be correlated across firewall (blocked connection) + endpoint (malware detected) + DNS (malicious domain queried) — automatically linked as one incident.

Q236 What is NSO service model?

Ans NSO service models abstract complex multi-device operations into simple service requests. Example: L3VPN service model. Engineer inputs: customer name, site locations, IP addressing. NSO translates to: BGP config on both PEs, VRF config, route-target, MPLS config. Service deployed consistently across all devices simultaneously. If service deleted: all device configs removed automatically. Benefits: huge time saving (one service request vs many manual CLI sessions), error elimination, consistency. NSO services written in YANG (data model) + Java/Python (device config logic). Used by major ISPs for automated service provisioning.

Q237 What is SuzieQ network observability?

Ans SuzieQ (open source) is a network observability tool for understanding network state. It: connects to devices via SSH or reads configuration files, stores operational state in database (routes, BGP sessions, OSPF neighbours, interfaces, MAC tables, LLDP), enables fast SQL-like queries across entire network: 'Show me all BGP sessions that are not Established across all routers', 'Find all routes in VLAN 100 routing table', 'Are all OSPF adjacencies full?'. Much faster than logging into each device. Supports: Cisco, Juniper, Arista, Cumulus, Palo Alto. Great complement to Ansible/Netmiko scripts.

Q238 What is configuration compliance checking?

Ans Compliance checking compares running device configurations against policy/template. Tools: NSO (compare running config vs service model), NAPALM (compare configs), custom Python (generate expected config from template, diff with actual config), DNA Center (compliance feature compares against golden template). Example policy: all routers must have NTP configured, syslog configured to specific server, enable secret set, no Telnet. Compliance scanner finds deviations and reports. Integration with ITSM: auto-ticket deviations. Essential for security audits (PCI-DSS, HIPAA, SOX compliance requires documented configuration standards).

Q239 What is ChatOps for network engineering?

Ans ChatOps integrates network operations into collaboration platforms (Slack, Microsoft Teams, Webex). Engineers type commands in chat: '@netbot show bgp 10.0.0.1' □ bot connects to router via Netmiko □ returns show bgp output in chat channel. Benefits: entire team sees what is being done (transparency), natural audit trail (chat history), faster response during incidents (no need to VPN, log in, type commands), collaborative troubleshooting. Common frameworks: Errbot (Python bot framework), Nautobot + ChatOps plugin, Cisco Webex bots. Reduces mean time to resolve (MTTR) during network incidents.

Q240 What is GitOps for networking?

Ans GitOps treats network configuration like software code — Git repository is the single source of truth for all network configuration. Workflow: all changes made as pull requests (PRs) to Git repo □ automated pipeline validates (syntax, policy compliance, Batfish simulation) □ peer review required □ merge triggers automatic deployment to network. Benefits: complete change history (who changed what when), easy rollback (git revert), consistent environments, automated testing prevents mistakes, multiple reviewers catch errors. Increasingly adopted in large enterprises and service providers as 'Network as Code'.

SECTION 09

Troubleshooting & Design

Q241 –
Q270

Methodology · Tools · Convergence · Design Principles · High Availability · Performance

Q241 What is the enterprise troubleshooting methodology?

Ans Structured troubleshooting process: 1) Define the problem precisely (what is broken, since when, who is affected, what changed?). 2) Collect information (show commands, logs, packet captures, user reports). 3) Identify possible causes (use OSI model — work up from Layer 1). 4) Create most likely hypothesis based on symptoms. 5) Test hypothesis (make one change). 6) Analyse results — did it fix the problem? 7) Identify root cause (not just symptom). 8) Document root cause, fix, and preventive measures. 9) Verify full system functionality. Single most important step: change ONE thing at a time.

TIP

Methodical troubleshooting = faster resolution + no new problems introduced. CCNP tests this!

Q242 How do you troubleshoot a slow network with no obvious outages?

Ans Performance troubleshooting: check interface error counters (input/output errors, drops — show interfaces). Check QoS queue drops (show policy-map interface — VoIP queue dropping?). Check CPU and memory (show processes cpu sorted, show memory statistics). Check routing table for unexpected paths (traceroute to destination, show ip route). Check for duplex mismatch (show interfaces — 'half-duplex' on a switch port = major performance issue). Check bandwidth utilisation (show interface — input/output rate). Packet capture: if application-level issue — capture at both ends, compare arrival of packets. Wireshark: look for TCP retransmissions (sign of loss or delay).

Q243 What is OSPF troubleshooting — neighbour stuck in states?

Ans OSPF troubleshooting by state: Stuck in ATTEMPT/DOWN: no Hello received — check connectivity (ping neighbour IP), check OSPF enabled on correct interface, check area number. Stuck in INIT: one-way communication — firewall or ACL blocking OSPF (protocol 89) from neighbour. Stuck in 2-WAY: normal for non-DR/BDR routers (only form full adjacency with DR/BDR). Check if both routers are DROTHER on multi-access segment. Stuck in EXSTART/EXCHANGE: MTU mismatch (show interfaces — both sides must have same MTU). ip ospf mtu-ignore to workaround. Stuck in LOADING: database exchange failing — clear ip ospf process.

Q244 What is BGP troubleshooting — route not appearing in routing table?

Ans BGP route troubleshooting: Step 1 — show bgp ipv4 unicast [prefix]: is prefix in BGP table at all? Not there: peer not sending it (check peer filters). There but not in routing table: look at '>' marker (no > = not best path). Why not best? Check: next-hop reachability (show ip route [next-hop-IP] — if not in IGP, BGP cannot use route). Check for inbound route-map filter. Step 2 — show bgp ipv4 unicast [prefix] detail: see all attributes (AS-path, LP, MED). Compare to alternative paths. Step 3 — show ip bgp neighbors [IP] received-routes vs advertised-routes: what is peer sending/what are we sending?

Q245 What is an OSPF LSA database overflow?

Ans LSDB overflow occurs when the OSPF database grows too large — typically in networks with too many external LSAs (Type 5) or too many routers. Symptoms: router memory exhaustion, high CPU during SPF, instability. Prevention: use stub areas to block Type 5 externals, summarise external routes at redistribution point, increase router memory. Detection: show ip ospf statistics (SPF run count growing?), show ip ospf database external (how many external LSAs?). OSPF database overflow protection: max-lsa [limit] (OSPF shuts down if LSA count exceeds limit). Emergency recovery: clear ip ospf process.

Q246 How do you troubleshoot EIGRP SIA (Stuck in Active)?

Ans EIGRP SIA troubleshooting: SIA means a router sent Queries but did not get Replies within 3 minutes. show ip eigrp topology active (see which routes are Active and which neighbours have not replied). To find which neighbour is slow to reply: show ip eigrp topology (look for 'Q' count — outstanding queries). Causes: query domain too large (fix: stub routing, summarisation at boundaries). Slow WAN link causing query timeout. Neighbour CPU overloaded (cannot process queries fast enough). Fix: add EIGRP stub on spoke routers. Summarise at distribution layer. Reduce hello/hold timers to detect failures faster.

Q247 How do you diagnose high CPU on a router?

Ans High CPU troubleshooting: show processes cpu sorted (top processes by CPU). Common culprits: IP Input (high: DDoS, large ACL, process switching), CEF (cef_bgp_update: BGP instability), Routing Protocols (OSPF/EIGRP instability — constant SPF runs), SNMP (too many frequent polls). Hardware switching issues: show platform resources (check forwarding engine health). Check for fast switching vs CEF: show ip cef (CEF should handle most traffic). Enable CPU threshold alerts: process cpu threshold type total rising 70 interval 30 (alert if CPU stays above 70% for 30 seconds). EEM can take automated action.

Q248 What is packet capture on IOS XE?

Ans Embedded Packet Capture (EPC) on IOS XE captures traffic without external tools. Configure: `ip access-list extended CAPTURE-ACL` □ `permit ip 192.168.1.5 any`. monitor capture CAP interface Gi0/1 both access-list CAPTURE-ACL (or: in/out only). monitor capture CAP buffer circular size 25 (25 MB circular buffer). monitor capture CAP start. Wait for traffic to be captured. monitor capture CAP stop. Export: monitor capture CAP export flash:capture.pcap. Analyse: copy to laptop and open in Wireshark. Alternatively: export while capturing to TFTP server. Very useful for validating ACLs, firewall policies, verifying QoS marking.

Q249 What is the three-tier vs two-tier vs spine-leaf design?

Ans Three-tier (traditional): Access (edge/user connectivity) □ Distribution (aggregation, policy) □ Core (high-speed backbone). Scalable for large campus. Slower east-west traffic (server-to-server). Two-tier (collapsed core): Access □ Core/Distribution collapsed. Simpler, lower cost, suitable for medium campus. Spine-leaf (data centre): every Leaf connects to every Spine (full mesh). Any server can reach any other server in 2 hops (Leaf □ Spine □ Leaf). Predictable latency, scales horizontally (add Spine for bandwidth, add Leaf for ports). ECMP across all Spine switches = maximum bandwidth. SD-Access uses a variation of this with SDA fabric.

Q250 What is the 80/20 rule in campus design?

Ans The old 80/20 rule: 80% of traffic stays local (within the department LAN), 20% crosses to other subnets. This favoured flat Layer 2 designs with one big VLAN per department — minimal routing needed. Modern reversal (20/80): 80% of traffic crosses subnets (server farms, cloud, internet — all outside the local segment). This justifies: routed access layer (Layer 3 to access switches), eliminating STP from access layer, optimising for north-south AND east-west traffic. SD-Access design assumes 20/80 traffic model — fabric provides any-to-any efficient routing.

Q251 What is network baseline and why is it important?

Ans A baseline is a documented snapshot of normal network performance: typical bandwidth utilisation (average and peaks), normal CPU/memory on devices, typical BGP/OSPF route counts, normal error rates, typical latency between locations. Uses: troubleshooting ('is this normal or abnormal?'), capacity planning (trend shows when to upgrade), SLA verification, change impact assessment. Tools: SNMP monitoring (Cacti, PRTG, Grafana), NetFlow analysis, DNA Center Assurance (tracks 30-day trends). Document baseline after any major change — new baseline after SD-WAN deployment.

Q252 What is STP root cause analysis?

Ans STP troubleshooting: sudden network outage affecting many users. Symptoms: show interface (high broadcast counter). show spanning-tree (unexpected root bridge or port states). Possible causes: new switch with lower Bridge ID plugged in (becoming Root Bridge — topology changed). PortFast port connected to switch (BPDU received — if no BPDU Guard, new neighbour topology). STP timer mismatch (misconfigured timers causing instability). Unidirectional link (UDLD or loop guard not configured). Recovery: identify unexpected Root Bridge, correct priorities, investigate new connections, verify BPDU Guard on all access ports.

Q253 What is QoS troubleshooting?

Ans QoS troubleshooting: VoIP quality issues (jitter, delay, drops): show policy-map interface Gi0/0 (check voice queue: 'drops' counter > 0 = VoIP being dropped — increase priority bandwidth % or check if non-voice traffic is mismarked as EF). Check trust boundary: show interface Gi0/0 (CoS marking trusted?). Verify DSCP marking end-to-end: IP phones mark EF, switch trusts CoS 5, router maps CoS 5 to DSCP 46. Verify QoS applied outbound on all WAN interfaces. show queue (see queue depths). Misclassified traffic: debug ip packet detail (check actual DSCP values in headers).

Q254 What is convergence time and how to improve it?

Ans Convergence time = time from failure detection to all routers having correct routing tables. Components: Detection (BFD vs hello timer: 50ms vs 40 seconds). Propagation (LSA or Update flooding: milliseconds in small network, seconds in large). Calculation (SPF: milliseconds for OSPF, DUAL for EIGRP). FIB download (forwarding table update: milliseconds). OSPF fast convergence: BFD + throttle timers + iSPF. EIGRP fast convergence: DUAL with FS (instant) + BFD. BGP fast convergence: BFD + route-reflector topology + additional paths (add-paths). Target: sub-second convergence for carrier-class networks.

Q255 What is ECMP and how do you verify it?

Ans ECMP (Equal-Cost Multi-Path) installs multiple equal-cost routes in the routing table and load balances traffic across them. OSPF: maximum-paths 16 (support up to 16 ECMP paths). EIGRP: maximum-paths 16. BGP: maximum-paths 4 (eBGP), maximum-paths ibgp 4 (iBGP). Verify: show ip route 192.168.1.0 (see multiple 'via' next-hops). show ip cef 192.168.1.0 detail (see per-flow load-sharing entries). Traffic distribution: CEF uses hash of source/destination IP (per-flow) by default. Per-packet load balancing: ip load-sharing per-packet (may cause out-of-order packets — avoid for TCP).

Q256 What is SLA monitoring for application performance?

Ans CCNP requires understanding application performance SLAs: VoIP: latency <150ms one-way, jitter <30ms, loss <1%. Video conferencing: latency <150ms, jitter <30ms, loss <0.1%. SAP/ERP transactions: latency <100ms response time. Tools: Cisco IP SLA (UDP jitter probes), ThousandEyes (cloud-based synthetic monitoring), DNA Center Assurance (client experience), AppDynamics (application performance monitoring). SD-WAN: built-in SLA monitoring — compares transport performance to thresholds, automatically reroutes if threshold violated.

Q257 What is BGP fast external fallover?

Ans BGP fast external fallover (enabled by default): when a directly connected eBGP peer's interface goes down, BGP immediately terminates the BGP session — does not wait for hold timer (3 minutes by default). This allows faster convergence to alternative path. Verify: `show ip bgp neighbors [IP] | include External`. When to disable: `bgp fast-external-fallover` (globally) or `no bgp fast-external-fallover` — only if you have multiple paths to eBGP peer through the same interface (unusual). Combined with BFD: even faster detection (50ms) + fast fallover = sub-second BGP convergence.

Q258 How do you design for high availability in campus?

Ans HA design principles: No single point of failure at any layer. Distribution: dual uplinks from every access switch to two different distribution switches. Core: redundant core switches (VSS, StackWise Virtual, or ECMP). Power: dual power supplies on critical switches, redundant UPS, generator backup. Protocols: RSTP (fast convergence), HSRP/VRRP (gateway redundancy with BFD tracking), LACP EtherChannel (uplink redundancy + bandwidth). Monitoring: SNMP traps for immediate failure notification, syslog centralisation, DNA Center Assurance. Documentation: accurate network diagrams, tested runbooks for common failure scenarios.

Q259 What is the difference between MPLS TE and BGP TE?

Ans MPLS Traffic Engineering (TE): controls exact path traffic takes through network hop-by-hop using RSVP LSPs. Very precise, supports bandwidth reservation, fast failover. Complex to deploy. BGP Traffic Engineering: influence routing using BGP attributes (AS-path prepending, communities, MED, LP). Controls which AS or exit point is used — not exact path within an AS. BGP TE is simpler but less precise. Combination: use BGP TE for coarse traffic direction (ISP A or ISP B), use MPLS TE within your network for fine-grained control. SD-WAN replaces MPLS TE in many enterprises — simpler application-aware routing without RSVP complexity.

Q260 What is the SD-WAN vs MPLS decision framework?

Ans Use MPLS when: guaranteed, measurable QoS is required (latency-sensitive financial trading, hospital VOIP), deterministic performance SLA is contractual, security within provider backbone is sufficient, legacy network requires MPLS interconnects. Use SD-WAN when: multiple WAN technologies needed (MPLS + internet + LTE), significant cost reduction required (50-80% savings), cloud-first strategy (DIA for SaaS), rapid branch expansion (ZTP), centralised management across hundreds of sites, application-aware routing automation needed. Hybrid: MPLS for voice/critical + SD-WAN for data/cloud — very common enterprise transition strategy.

Q261 How do you troubleshoot a BGP route not advertised to peer?

Ans Route not advertised troubleshooting: Step 1 — show bgp ipv4 unicast neighbors [IP] advertised-routes: see what is actually being advertised. Not there? Step 2 — Is route in BGP table? show bgp ipv4 unicast [prefix]. Not in BGP: is it being originated? (network command or redistribute). In BGP but not advertised: route-map filtering it? (show bgp ipv4 unicast [prefix] detail — see outbound policy). Synchronisation enabled and no IGP route? Step 3 — clear ip bgp [IP] soft out: regenerate advertisements (picks up new policies). Step 4 — Is peer established? Prefix limit reached? (show bgp neighbor — check maximum-prefix).

Q262 What is dual-homed BGP and how do you manage it?

Ans Dual-homed: your network connected to two different ISPs. Goals: redundancy (one ISP fails, other continues), load balancing (use both ISPs simultaneously). Challenges: inbound control (which ISP sends you traffic), outbound control (which ISP you use for each destination). Inbound: AS-path prepending (longer path = avoid that ISP). MED (suggest preferred entry point to same ISP). Outbound: Local Preference (prefer routes from ISP-A over ISP-B). Weight (per-router preference). prefix-based policies (route specific destinations via specific ISP). Testing: withdraw one ISP connection and verify traffic fails over.

Q263 What is Cisco hierarchical network design principles?

Ans Cisco three-layer hierarchy with specific functions: Access Layer: end-user connectivity (switch ports for PCs/phones/APs), port security, 802.1X, PoE, QoS marking/trusting. Distribution Layer: routing between VLANs, policy enforcement (ACLs), summarisation of routes toward core, redundancy with HSRP/ECMP. Core Layer: high-speed backbone, no packet manipulation (no ACLs, no routing decisions) — just fast forwarding, redundancy, maximum uptime. Design rules: core should not be configuration boundary, distribution handles policy, access handles user features. Modular design: each layer can be scaled independently.

Q264 What is BFD configuration and verification?

Ans BFD configuration: interface level: `bfd interval 200 min_rx 200 multiplier 3` (send every 200ms, expect reply within 200ms, declare down after 3 misses = 600ms detection). Protocol integration: `ip ospf bfd` or `neighbor [IP] fall-over bfd` (BGP). Global BFD: `router ospf 1` `bfd all-interfaces`. Verification: `show bfd neighbors detail` (see session state, timers, missed count). `show bfd neighbors statistics` (packet counts per session). Common issues: BFD not supported on sub-interfaces or tunnel interfaces on some platforms, MTU issues preventing BFD packets. One BFD session per IP pair — shared by OSPF, EIGRP, BGP simultaneously.

Q265 What is FHRP gateway tracking best practice?

Ans FHRP tracking best practice: track the uplink toward the internet (not just the direct interface). Example: HSRP Active on R1 with ISP uplink. Track IP SLA to 8.8.8.8 (or track ISP interface). If ISP link or route to internet fails: tracking object goes down $\square$ HSRP priority decremented (R1 drops from 200 to 130) $\square$ R2 (priority 150) becomes Active $\square$ clients now use R2 with working internet path. Without tracking: R1 stays HSRP Active even after losing internet — all client traffic black-holed. Deploy: `ip sla` $\square$ `track object` $\square$ `standby track`. Test: shut ISP interface, verify HSRP preempts.

Q266 What is campus wireless design — AP placement?

Ans AP placement principles: Desired coverage: -65 to -67 dBm at cell edge. Cell overlap: 15-20% overlap between adjacent cells (for seamless roaming). Dense environments: smaller cells (lower power) — more APs per area. Open spaces: larger cells (higher power) — fewer APs. Avoid: microwaves (interference), metal structures (reflection), thick concrete walls (absorption). 5 GHz: provides capacity, shorter range — place APs every 20-30 feet in office. 2.4 GHz: provides coverage, longer range. Best practice: separate 2.4 and 5 GHz antennas, use WLC band steering, maintain 3 AP overlap at any client location for roaming.

Q267 What is the impact of STP on campus design?

Ans STP problems in campus: Convergence time (30-50 sec for classic STP, 1-2 sec RSTP — traffic loss during failover). Topology changes cause MAC table flush (temporary flooding — performance impact). STP instability (flapping links cause constant TCNs). Layer 2 loops kill networks instantly if STP fails. Modern approach: eliminate STP where possible. Routed access: each access port is a Layer 3 routed link — no STP between access and distribution. SD-Access: STP completely eliminated (VXLAN overlay with IP underlay). Where STP must run: within VLANs on EtherChannel between distribution switches. Rule: reduce STP domain size to minimum possible.

Q268 What is capacity planning for WAN links?

Ans WAN capacity planning: gather data: NetFlow or SNMP bandwidth utilisation (average + peak). application breakdown (which applications use most bandwidth). growth projections (20-30% annual growth is typical). Plan for: maximum utilisation < 70% of link capacity (leave headroom for peaks and QoS). VoIP: each call = ~100 Kbps (G.711) or ~27 Kbps (G.729 compressed). Plan bandwidth per call + headroom. Factor in backup: if primary link fails, secondary must handle full traffic load. SD-WAN: use application performance data from vManage to right-size links — not just utilisation.

Q269 What is the impact of large routing tables on router performance?

Ans Large routing tables (especially full BGP table ~900,000 routes) impact: Memory: each route entry consumes memory — BGP full table requires 1-2 GB RAM. CPU: SPF/DUALIAL recalculation time increases with table size. FIB download: pushing large forwarding table to line card hardware takes longer. CEF switching: large FIB with many entries = longer lookup (though hardware TCAM makes this microseconds). Solutions: hardware-based CEF (no CPU involvement for forwarding), route summarisation (reduce internal table size), BGP route filtering (only accept needed prefixes from ISP), dedicated high-memory BGP routers at internet edge.

Q270 What is the impact of network design on OSPF convergence?

Ans OSPF design for fast convergence: fewer routers per area (smaller SPF tree = faster calculation). Proper use of stub areas (fewer LSAs). Summarisation at ABRs (failures in one area do not cause SPF in others). Limit LSA types: stub areas block Type 5, totally stub blocks Type 3+5. Physical design: redundant links with BFD (fast failure detection). Point-to-point interfaces (no DR/BDR election = faster adjacency). iSPF (partial SPF recalculation). SPF throttle (tuned per network — lower timers for fast convergence but more CPU). Network diameter: longer paths = more routers informed = slower convergence. Keep OSPF areas compact.

SECTION 10

CCNP Career & Exam Preparation

Q271 –
Q300

350-401 ENCOR · 300-410 ENARSI · Exam Strategy · Career Paths · CCIE · Labs

Q271 What is the CCNP Enterprise certification and what exams are required?

Ans CCNP Enterprise (Cisco Certified Network Professional Enterprise) requires two exams: 350-401 ENCOR (Enterprise Core): core exam covering Infrastructure (routing, switching, wireless, automation), Network Assurance, Security, and Automation. 120 questions, 120 minutes. 300-410 ENARSI (Enterprise Advanced Routing and Services): concentration exam covering advanced routing (OSPF, EIGRP, BGP, redistribution), infrastructure services (DMVPN, IPsec, VRF), and infrastructure security. You choose ONE concentration exam — ENARSI is most popular for network engineers.

TIP*ENCOR + ENARSI = CCNP Enterprise. Also qualifies for CCIE Enterprise written exam!***Q272** What topics are in the ENCOR 350-401 exam?

Ans ENCOR 350-401 domains and weightings: Architecture (15%): campus/WAN/cloud/SD-Access/SD-WAN design. Virtualisation (10%): device virtualisation, VRF, VxLAN. Infrastructure (30%): switching (STP, EtherChannel), routing (OSPF, EIGRP, BGP, redistribution, path control). Network Assurance (10%): Cisco DNA Center, streaming telemetry, SNMP, NetFlow, IP SLA. Security (20%): device hardening, AAA, ZBFW, 802.1X, DMVPN, IPsec, TrustSec. Automation (15%): Python, Ansible, RESTCONF, NETCONF, model-driven programmability. Study proportionally to domain weights.

Q273 What topics are in the ENARSI 300-410 exam?

Ans ENARSI 300-410 domains: Layer 3 Technologies (40%): OSPF multi-area, EIGRP advanced, BGP (attributes, filtering, redistribution), redistribution between protocols, path control (policy-based routing), route summarisation. VPN Technologies (20%): DMVPN Phase 1/2/3, IPsec, FlexVPN, GRE. Infrastructure Services (25%): DHCP (server, relay, snooping), NTP, SNMP, NetFlow, IP SLA, AAA (RADIUS, TACACS+), FHRP. Infrastructure Security (15%): device hardening, ACLs, uRPF, prefix filtering, route filtering. ENARSI is heavily lab-based — practice every technology.

Q274 What is the best study strategy for CCNP Enterprise?

Ans Proven CCNP study approach: Official cert guides (OCG) by Wendell Odom for ENCOR, Narbik Kocharians ENARSI guide. Video courses: INE (excellent lab-heavy content), CBT Nuggets, Jeremy's IT Lab (ENCOR). Lab practice: absolutely essential — CCNP cannot be passed by reading alone. Cisco CML or EVE-NG for labs. Focus on configuring, verifying, and troubleshooting each technology. Practice exams: Boson ExSim (most accurate). Target 85%+ in practice before scheduling real exam. Exam day: read questions carefully — CCNP questions often have subtle wording differences.

Q275 What career opportunities does CCNP Enterprise open?

Ans CCNP Enterprise roles: Senior Network Engineer (design and implement campus/WAN/cloud networks), Network Architect (design complex multi-site networks), Network Operations Lead (manage NOC team, escalation point), Enterprise Wireless Engineer, SD-WAN specialist, Network Automation Engineer. Salary India: CCNP certified: ₹8-20 LPA depending on experience. Senior roles with CCNP + 5 years: ₹15-30 LPA. Large IT services, banks, telecom, FAANG India operations. US: \$80,000-\$130,000 for CCNP-level roles. CCNP is a significant salary differentiator from CCNA.

TIP

Your 4+ years experience + CCNP = senior network engineering roles are accessible now!

Q276 What is CCIE Enterprise and how does it build on CCNP?

Ans CCIE Enterprise (Cisco Certified Internetwork Expert) is the highest Cisco certification. Requires: ENCOR 350-401 written exam (same as CCNP ENCOR — passing it satisfies CCIE written), 8-hour practical lab exam (at Cisco testing labs). Lab exam: CCIE Enterprise Infrastructure or CCIE Enterprise Wireless. Tests: complex multi-technology design, configuration, and troubleshooting with no documentation. Pass rate: approximately 25-30% first attempt. CCIE holders: India ₹30-60 LPA+, USA \$120,000-\$200,000+. Typically requires 3-5 years deep hands-on experience after CCNP. Elite certification.

Q277 What is Policy-Based Routing (PBR)?

Ans PBR routes packets based on source IP, DSCP marking, or packet length — not just destination. Use cases: route VoIP traffic through low-latency ISP regardless of routing table, route management traffic through dedicated management path, different users get different internet gateways. Configure: `ip access-list extended PBR-MATCH` `permit tcp 192.168.1.0 0.0.0.255 any eq 80`. `route-map PBR-MAP` `permit 10` `match ip address PBR-MATCH` `set ip next-hop 10.0.0.1`. Apply: `interface Gi0/1` `ip policy route-map PBR-MAP`. Verify: `show ip policy`, `debug ip policy`.

Q278 What is MPLS LDP configuration and verification?

Ans Configure MPLS with LDP: `mpls ip` (enable MPLS globally). `mpls ldp router-id Loopback0 force`. `interface Gi0/0` `mpls ip` (enable LDP on interface). LDP uses IGP (OSPF or IS-IS) for neighbour discovery. Verify: `show mpls ldp neighbor` (see LDP peer relationships). `show mpls ldp bindings` (see label database — local and remote bindings per prefix). `show mpls forwarding-table` (see label switching table — incoming label, outgoing label, interface). `ping mpls ipv4 10.0.0.1/32` (test LSP). Troubleshoot: LDP not forming — check routing (can reach peer IP?), check `mpls ip` on interface.

Q279 What is OSPFv3 address-family configuration?

Ans OSPFv3 address-family model supports both IPv4 and IPv6 in one process: `router ospfv3 1` `address-family ipv4 unicast` `router-id 10.0.0.1` `passive-interface Gi0/2` `address-family ipv6 unicast` `router-id 10.0.0.1`. On interface: `interface Gi0/0` `ospfv3 1 ipv4 area 0` `ospfv3 1 ipv6 area 0`. Verify: `show ospfv3 1 ipv4 neighbor`, `show ospfv3 1 ipv6 neighbor`. More complex than separate OSPFv2 + OSPFv3 processes but cleaner when managing both IPv4 and IPv6 in same OSPF domain.

Q280 What is BGP best-path logging?

Ans BGP logs best path changes when the preferred route changes. Enable: `bgp log-neighbor-changes` (logs when BGP sessions go up/down). `debug ip bgp [IP]` events (lightweight debug for specific peer — shows state changes, route updates). To log specific best-path changes: `bgp bestpath compare-routerid` + `bgp log-neighbor-changes` provides some visibility. For detailed route tracking: `ip sla` + `track` + event routing to detect specific prefix availability changes. In production: do not run debug — use syslog server to capture bgp neighbor change messages automatically.

Q281 What is VRF-Lite?

Ans VRF-Lite uses VRF without MPLS — just multiple routing tables on a single router, shared with CE devices via standard routing (OSPF, EIGRP, BGP, or static). Use cases: enterprise network segmentation (separate routing domain per department without MPLS), managed service provider using a single physical router to serve multiple customers with overlapping addresses, network virtualisation in data centres. Configure: `ip vrf DEPT-A` `rd 1:1`. Interface: `ip vrf forwarding DEPT-A` `ip address [address]`. Routing: `router ospf 100 vrf DEPT-A` (per-VRF routing process). Verify: `show ip route vrf DEPT-A`.

Q282 What is the ENCOR lab topology to practice?

Ans Essential CCNP ENCOR lab topology: 3-4 routers in a ring/partial mesh for multi-area OSPF and BGP. 2 core switches + 4 access switches (RSTP, EtherChannel, VLANs, inter-VLAN routing). WLC + 1-2 APs (or simulated) for wireless section. Firewall (ASA or ZBFW router) for security section. Scenarios to practice: OSPF + EIGRP redistribution with route tagging, BGP with full attribute manipulation, EtherChannel with LACP + BPDU Guard, DMVPN Phase 1/2/3, QoS with LLQ + CBWFQ, ZBFW + IPsec VPN. Cisco CML or EVE-NG with real IOS XE images.

Q283 What additional certifications complement CCNP Enterprise?

Ans Complementary certifications: Cisco DevNet Associate (DEVASC 200-901): network automation — highly recommended alongside CCNP. CompTIA Security+ or CISSP: for security roles. AWS Solutions Architect: cloud networking knowledge. Azure Network Engineer: for Microsoft cloud focus. Cisco CCNP Security: if moving toward security architecture. Juniper JNCIS-ENT: for multi-vendor environments. ITIL Foundation: service management. CKA (Certified Kubernetes Administrator): if networking data centre containers. CCNP + DevNet + cloud certification = complete modern network architect skillset.

Q284 What are the CCNP-level troubleshooting commands to master?

Ans Critical CCNP commands: `show ip ospf neighbor detail`, `show ip ospf database`, `show ip ospf statistics`, `show bgp ipv4 unicast summary`, `show bgp ipv4 unicast neighbors [IP] advertised-routes/received-routes`, `show bgp ipv4 unicast [prefix] detail`, `show eigrp address-family ipv4 topology`, `show ip route [prefix] longer-prefixes`, `show ip cef [prefix] detail`, `show policy-map interface [int]`, `show spanning-tree vlan [id] detail`, `show interfaces [int] (errors, duplex, speed)`, `show ip nat translations`, `show crypto session`, `show bfd neighbors detail`, `show track [object]`.

Q285 What is the CCNP Enterprise exam recertification policy?

Ans CCNP Enterprise is valid for 3 years. Recertification options: Pass any CCNP concentration exam (300-4xx). Pass CCIE written exam (350-401 ENCOR qualifies for CCIE and recertifies CCNP simultaneously). Pass any Cisco specialist exam. Earn continuing education (CE) credits (80 CE credits for CCNP). Pass any higher-certification exam (CCIE written). Cisco's Learning Partner exams: some approved training courses provide CE credits. Track status at certificationtracker.cisco.com. Note: CCNA certification automatically recertified when you pass CCNP.

Q286 What are common CCNP exam mistakes to avoid?

Ans Common traps: Confusing BGP path selection order (Weight vs LP vs AS-path vs MED — know the exact order). OSPF LSA types — which router generates which type. RSTP vs STP port roles (Alternate vs Backup vs Root port). EIGRP Feasibility Condition (RD must be less than FD of Successor — not less than or equal). OSPF network types — which ones require manual neighbour config. BGP next-hop issue (iBGP peers cannot reach eBGP peer address). Zone-based firewall — traffic within same zone allowed by default, between zones denied by default. Always read questions twice — subtle wording differences change the answer.

Q287 What is practical automation skill for CCNP candidates?

Ans Practical automation for CCNP: Write a Python script using Netmiko to connect to 10 routers and collect show version output. Write a script to check if all BGP sessions are in Established state — alert if any are not. Use RESTCONF to configure a loopback interface on IOS XE router. Write an Ansible playbook to configure NTP server on all routers. Use DNA Center API to get list of all devices and their health score. Parse show ip ospf neighbor with textFSM or Genie. These practical skills demonstrate automation competency for both CCNP exam and job interviews.

Q288 What is the industry trend toward network as code?

Ans Network as Code (NaC): manage entire network infrastructure through code, version control, and automated pipelines — same discipline as software development. Tools: Git (version control for configs), Ansible/Terraform (automated deployment), Batfish (pre-deployment validation), pyATS (post-deployment testing), DNA Center API (Cisco campus), vManage API (SD-WAN). Result: every network change is: reviewed (pull request), tested automatically, deployed consistently, tracked historically, and easily reversible. Companies like Microsoft, Google, and large ISPs fully implement NaC. This is the direction for the next decade of networking.

Q289 What is the difference between CCNP and CCIE level knowledge?

Ans CCNP level: understand technologies conceptually and operationally, configure and troubleshoot common scenarios, know which technology to use for which problem, handle real-world network issues. CCIE level: deep protocol understanding (byte-level packet formats, algorithm specifics), configure complex multi-technology scenarios under time pressure (8-hour lab with no documentation), troubleshoot without hints in time-boxed scenarios, optimise for both correctness and efficiency, design complete network architectures. CCNP is enterprise competency; CCIE is expert mastery. Path: CCNA □ CCNP (2-3 years experience) □ CCIE (3-5 more years of deep experience).

Q290 What is the value of CCNP in the enterprise job market?

Ans CCNP Enterprise market value: Most enterprise network engineer job postings require CCNP or equivalent. Differentiates from CCNA (shows advanced skills). Respected in: banks, hospitals, government, telecom, IT services companies. Cisco partner companies: need CCNP-certified staff for Cisco partner status. Consulting firms: senior consultant roles require CCNP minimum. Defence and government: some positions require Cisco certifications specifically. Cloud transition: CCNP + cloud cert (AWS/Azure) = hybrid cloud network engineer — very high demand. AI networking (Cisco Catalyst AI, Meraki AI): next frontier for CCNP engineers.

Q291 What is ENCOR domain — Automation and Programmability?

Ans Automation domain topics: Python basics (variables, loops, functions, data structures, JSON parsing). Ansible (playbooks, inventory, modules for Cisco IOS/NX-OS). NETCONF (XML, operations, ncclient library). RESTCONF (HTTP methods, JSON, requests library). Cisco-specific APIs (DNA Center REST API, vManage REST API). IOS XE model-driven programmability (YANG, NETCONF, RESTCONF, gRPC telemetry). Guestshell and EEM. Git basics. Data formats (JSON vs XML vs YAML). You do not need to write complex code — understand syntax, read scripts, identify correct API calls.

Q292 What is ENCOR domain — Network Assurance?

Ans Network Assurance domain: DNA Center Assurance (what it monitors, how to interpret health scores). SNMP (v3 configuration, traps vs polls). IP SLA (configure UDP jitter probe, interpret statistics). NetFlow/Flexible NetFlow (configure and verify). Syslog (configure logging levels, send to server). Cisco IOS IP SLA tracking (object tracking with routes and HSRP). Streaming telemetry (model-driven telemetry concepts, gRPC, gNMI). Debug best practices (targeted debug, EPC packet capture). Using show commands for verification. DNA Center features for network health visibility.

Q293 What is the Cisco DevNet Associate (DEVASC) exam?

Ans DEVASC 200-901 covers: Software Development (APIs, Python, Git, testing). Network Fundamentals (protocols, automation concepts). Application Deployment (Docker, Kubernetes basics). Infrastructure and Automation (Ansible, Terraform, CI/CD). Cisco Platforms and Development (DNA Center API, Meraki API, Webex API, ACI API, SD-WAN API, IOS XE programmability). Impact (network, security, collaboration, data centre automation use cases). Pairs perfectly with CCNP Enterprise — CCNP provides network knowledge, DEVASC provides programming. The combination creates Network DevOps engineers.

Q294 What is ENAUTO 300-435 (Enterprise Automation)?

Ans ENAUTO 300-435 (Automating and Programming Cisco Enterprise Solutions) is the CCNP automation specialist exam. Topics: Network Programmability Foundation (Python, REST API, NETCONF, YANG, model-driven telemetry). Cisco Platforms (DNA Center API, vManage SD-WAN API, Meraki API, Cisco NSO, IOS XE programmability). AppDynamics, ThousandEyes for monitoring. CI/CD pipelines for network automation (Git, Jenkins, Ansible). Network programmability design (when to use which tool). This is the most valuable specialist exam for modern network engineers pursuing automation careers.

Q295 How does CCNP Enterprise knowledge apply to cloud networking?

Ans CCNP skills directly transfer to cloud: BGP: AWS, Azure, GCP all use BGP for connectivity (Direct Connect, ExpressRoute, Cloud Interconnect). OSPF/routing: internal cloud VPC routing concepts same as on-premises. VPN: site-to-site IPsec to cloud (same concepts as on-premises). QoS: AWS Direct Connect QoS, Azure ExpressRoute bandwidth allocation. SD-WAN: extends directly to cloud (Cisco cloud onramp). Automation: cloud APIs (AWS CLI, Azure CLI, Terraform) — same REST/API concepts. Security: cloud security groups = simplified ACLs. Network engineer with CCNP + cloud cert: highest market demand.

Q296 What are the newest features in IOS XE 17.x relevant to CCNP?

Ans Recent IOS XE 17.x relevant features: Catalyst 9000 enhanced switching performance, SD-Access 2.0 improvements (scalability, simplified underlay). IOS XE 17.x native telemetry improvements (gRPC efficiency). OpenConfig YANG model support expanded. Dual-stack routing improvements (IPv4+IPv6 convergence). BGP additional paths (ADD-PATHS) — CCNP now includes this. Enhanced VXLAN/EVPN support. FlexVPN improvements (IKEv2). DNA Center 2.x: new AI-driven insights, enhanced fabric provisioning. Always check Cisco's release notes and exam blueprint for latest topics included in current ENCOR exam version.

Q297 What is Cisco ThousandEyes?

Ans ThousandEyes (acquired by Cisco 2020) is a cloud-based network intelligence platform providing end-to-end visibility across internet, cloud, and enterprise networks. Features: active monitoring from global vantage points (internet agents), end-user experience monitoring (browser test, HTTP test), BGP route visibility (see global BGP changes in real time), network path visualisation (traceroute from multiple global points simultaneously), alert when internet path degrades. Enterprise agents: run on-premise or on employee laptops — monitor performance from user's perspective. Integration: DNA Center, ISE, SD-WAN vManage.

Q298 What is the role of CCNP engineer in a DevOps/Agile environment?

Ans Modern CCNP engineers in DevOps organisations: write infrastructure as code (Terraform for cloud networking, Ansible for on-premises), participate in sprint planning for network changes (not weeks of lead time — hours or days), use Git for all configuration changes with peer review, maintain CI/CD pipelines for network automation, collaborate with developers (network policies that enable application deployment), use SRE (Site Reliability Engineering) practices (SLOs, error budgets, postmortems for outages). Soft skills equally important: communication with dev teams, documentation culture, blameless incident review.

Q299 What study resources are most valuable for CCNP?

Ans Top CCNP study resources: Cisco Press Official Cert Guides (OCG): authoritative, complete coverage. INE training platform: best lab-heavy video content for CCNP/CCIE. Cisco DevNet Learning Labs: free hands-on automation labs. Cisco CML or EVE-NG: lab practice with real IOS XE. Boson ExSim: most realistic practice exams (highly recommended). Cisco documentation ([cisco.com/c/en/us/support](https://www.cisco.com/c/en/us/support)): authoritative command reference. Reddit r/ccnp: study tips, recent exam feedback (no brain dumps). YouTube: Jeremy's IT Lab (ENCOR), David Bombal (automation). Network Collective podcast: industry perspective and advanced topics.

Q300 What final advice do you have for CCNP Enterprise success?

Ans From one network engineer to another: The CCNP is achievable with the right approach — it rewards depth, not breadth. The exam tests whether you can APPLY knowledge, not just recall it. Lab practice is non-negotiable — if you cannot configure and troubleshoot each technology from memory, you are not ready. Focus on your weakest domains (for most people: BGP attributes and automation). Use your real-world experience: connect CCNP topics to actual problems you have solved. Your FortiGate and network security background directly applies to ZBFW and security sections. CCNP is the credential that opens architect-level doors. Invest the time — it is absolutely worth it.

TIP

Akhilesh, your 4+ years with FortiGate, VPN, DHCP, and Active Directory is a massive advantage for CCNP. You are already practising half the exam topics daily. Go for it!